Shadow Aware Object Detection and Vehicle Identification via License Plate Recognition

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ABSTRACT

SHADOW AWARE OBJECT DETECTION AND VEHICLE IDENTIFICATION VIA LICENSE PLATE RECOGNITION

This research proposes a comparative study between three shadow removal algorithms and their application in license plate recognition. The idea is to monitor a junction for red light violators and when one is detected to capture an image of the vehicle along with its license plate details, which could be used to identify the driver. Therefore, the focus of this research is on foreground segmentation, moving shadow detection and elimination and license plate recognition procedures.

Moving cast shadows need careful consideration in order to develop accurate object detection. Shadows may be misclassified as part of the foreground object and this at times could cause merging of foreground objects, object shape distortion, and even object losses (due to the shadow cast over another object). Therefore, the removal of the shadows aids in more accurate detection of vehicle and hence, a correct foreground for license plate recognition algorithms.

In this thesis, a background estimation / subtraction technique is applied to segment the foreground and then three different shadow detection and removal techniques are implemented and compared. First technique is based on the cast shadow observations in luminance chrominance and gradient density making use of a combined probability map called Shadow Confidence Score (*SCS*). Second method exploits the *HSV* color space transform to converts pixel information from *RGB* color

space to *HSV* domain. Third method is a hybrid color and texture-based approach where chromacity conditions and texture similarities or dissimilarities of input and background frames are considered in order to detect cast shadow parts. To evaluate the performance of the various shadow removal algorithms ground-truth video frames have been used as a quantitative scale. Finally, a correlation based *LPR* algorithm was used to recognize the plate number of red light violators. First the Radon transform was applied to estimate the skew angle of the detected foreground objects and rotation angles were corrected. Then a color and edge based localization for the license plate was carried out. After localization of the plate the individual characters were segmented out using connected component analysis and the characters to test were separated based on their Euler numbers. This was done because Euler number filtering before the character recognition procedure is known to improve the accuracy of the character recognition and also at the same time speeds up the processing.

Keywords: Background estimation, foreground segmentation, shadow removal, skew correction, license plate recognition,

ÖZET

GÖLGE BİLİNÇLİ NESNE TESBİTİ VE PLAKA TANIMA YOLUYLA ARAÇ TANIMLAMA

Bu araştırma üç farklı gölge kaldırma yaklaşımı arasında karşılaştırmalı bir çalışmayı ve bu çalışmanın görüntülü şehir trafik izlemede kırmızı ışık ihlali tesbiti durumunda araç plakası tanımlama amacıyla bir kavşaktakta trafik kurallarını ihlal eden araçların resimlerinin plakala detaylarıyla beraber yakalanmasının uygulamasını sunmaktadır. Bu araştırmanın odağı önplan bölütleme, hareketli gölge tesbiti ve giderilmesi ve plaka tanımlamadır.

Hareketli gölgelerin düşüm noktasının doğru nesne tebiti için dikkatle gözönünde bulundurulması gerekmektedir. Gölgeler önplanın bir parçası olarak yanlış sınıflandırılabilirler ve bu önplan nesnelerinin karışmasına, nesne şeklinin bozukluğuna, ve hatta nesne kaybına (gölgenin birbaşka nesne üzerine düşmesi sebebiyle) sebep olacaktır. Bu yüzden, gölgelerin kaldırılması araçların daha kesin tesbit edilebilmesine ve dolayısıyla araç plaka tesbiti için daha uygun önplan oluşturulmasına yardım etmektedir.

Bu tezde, önplanı bölütlemek amaçlı arkaplan tahmin/çıkarma tekniği uygulanmıştır ve sonra üç farklı gölge tesbit ve kaldırma yaklaşımı uygulanmış ve karşılaştırılmıştır. İlk yaklaşım, gölge güvenlirlik hesabı diye adlandırılan birleşik olasılık haritası kullanılarak parlaklık/renklilik ve meyil yoğunluğundaki gölge düşüm gözlemlerine dayanmaktadır. İkinci yöntem piksel bilgisini *RGB* renk uzayından *HSV* alanına çevirmek için *HSV* renk boşluk transformunu kullanır.

Üçüncü yöntem parlaklık durumlarının ve doku benzerliklerinin ya da girdi ve arkaplan farklılıklarının gölge düşüm parçalarını tesbit etmek amacıyla göz önünde bulundurulduğu parlaklık ve doku tabanlı yaklaşımdır. Çeşitli gölge kaldırma algoritmalarının performansını ölçmek için referans video kareleri niceliksel ölçü olarak kullanılmıştır. Son olarak, plaka tanımlama ve plaka numarası tanıma için bir algoritma kullanılmıştır.

Anahtar Kelimeler: arkaplan kestirimi, önplan bölütleme, gölge kaldırma, yamukluk açisi duzeltme, arac plakası tanime

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LIST OF ABBREVIATIONS / SYMBOLS

YC_bC_r	Luminance; Chroma: Blue; Chroma: Red
SCS	Shadow Confidence Score
MFM	Moving Foreground Mask
RGB	Red, Green, Blue
GD	Gradient Density
SAKBOT	Statistical and Knowledge Based Object Tracker
HSV	Hue, Saturation, Value
CMY	Cyan, Magenta, Yellow
YUV	Luminance, Bandwidth, Chrominance
BD	Brightness Distortion
CD	Chromacity (Color) Distortion
LPR	License Plate Recognition
CCA	Connected Component Analysis
WHR	Width to Height Ratio
MATLAB	MATrix LABoratory
GBH	Group Based histogram
GMM	Gaussian Mixture Model
TRNC	Turkish Republic Of North Cyprus
ISS	Intelligent Surveillance System
SNP	Statistical Non-Parametric
SP	Statistical Parameter
DM	Deterministic Model-Based

DNM	Deterministic Non-Model-Based
SVDD	Support Vector Domain Description
FNN	Feed-forward Neural Network
BRLS	Block Recursive Least Square
MSE	Mean Square Error
SVM	Support Vector Machine
CCL	Connected Component Labeling
SOM	Self Organizing Map
HMM	Hidden Markov Model

CHAPTER 1

INTRODUCTION

Real time segmentation of dynamic regions or objects in video or image sequences is often referred to as "*background subtraction*" or "*foreground segmentation*" and is a fundamental step in many computer vision applications. Some examples include automated visual surveillance, traffic flow calculations, object tracking, and detection of red light violations.

With new developments in computer and communications technologies the need for improving Intelligent Surveillance Systems (*ISS)* technologies is becoming more significant. The importance of visual traffic surveillance is its role in capturing traffic data, detecting accidents and in safety management in general. It has been demonstrated that vision based information processing will result in improved operational efficiency.

The very first step in visual traffic processing is the segmentation of mobile objects in image sequences. Based on the recorded image sequences a background can be estimated and an old technique known as background subtraction is applied to segment moving objects from each frame of the video. Background subtraction has been utilized with various background estimation algorithms for different traffic scenes. When the estimated background is fine then subtraction would lead to a good estimate of the foreground mask [1]. However if the estimate is not good enough, then the background subtraction method may result in a rough approximate of the moving region. Furthermore with slow moving traffic it may even fail to provide a result.

Illumination changes, shadow and inter-reflections, background fluctuations, and crowded scenes are phenomena which cause problems for background estimation. To handle some of these problems computationally expensive methods are employed, where in practice such processing has to run in real-time. For instance, most vision-based surveillance systems collect videos that are to be analyzed offline at some later time. Background subtraction methods used in this manner, need also to place a record of the date on the video so they can report on the results using this date [3].

Real time methods are not able to handle properly one or more common phenomena, such as global illumination changes, shadows, inter-reflections, similarity of foreground object colors to that of the background, and non-static backgrounds (i.e., tree branches and leaves waving in the wind). Different background subtraction methods have been proposed over the last decades. The simplest class of methods uses color or intensity as input and employs background feature values (stillness) at each pixel to produce an independent, uni-modal distribution. For example, Single-Gaussian (SG) model and group-based histogram background estimation are such methods. [3]

When the current input frame is significantly different from the distribution of expected background pixel vector, a foreground is detected on a per-pixel basis. While dealing with color or intensity based background estimation algorithms shadow points can also be detected as a part of the extracted foreground. In such cases one needs to apply shadow detection and removal algorithms to have a more correct *FG* representation.

Shadow is the name of the region produced by partial or entire occlusion of direct light from a light source by an object. The procedure for identifying shadows is divided into three processes: low level, medium level and high level [4]. The low level process detects regions which are darker than their surroundings. Shadows are among the dark regions. A middle level process detects features in dark regions such as the penumbra, self-shadows and cast shadows. The object regions are adjacent to dark regions. A high level process integrates these hypotheses and confirms the consistency among the light directions estimated from them [4]. In general, a shadow region is divided in to two parts: the self shadow and the cast shadow. The self-shadow is the part of the object which is not illuminated by direct light. The cast shadow is the area projected by the object in the way of direct light. Cast shadows in the real world belong to illumination effects, because the light ray on its way from the light source is affected by more than only one reflection on object surface.

Umbra is the part of a cast shadow where direct light is totally blocked by its object. On the other hand, Penumbra is a part of a cast shadow where direct light is partially blocked. These parts are depicted in Figure1.1. A point light source only generates umbra in shadows. An “area” light source generates both penumbra and umbra. When a penumbra is very small, it may not appear in an image due to digitizing effects.

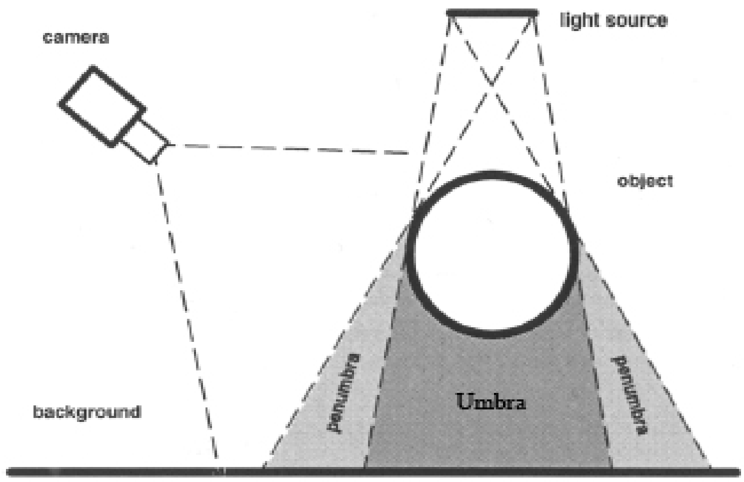


Figure 1. 1: Cast shadow parts: "umbra" and "penumbra"

In comparison to the penumbra area, umbra has lower light intensities because umbra is receiving no light from the light source. These intensities increase gradually from umbra to penumbra. The calculation of the luminance in a penumbra is similar to that of an object surface, except that only a partial light source needs to be considered [4]. The variations of the intensities in the penumbra area are not a simple function of a light source and object. It is extremely difficult to achieve theoretical formula of the intensities in a penumbra for an arbitrary object and an arbitrary light.

As declared before, moving vehicles are often extracted with their associated cast shadows after the application of background subtraction to traffic image sequences. This phenomenon may lead to object losses, object shape distortion of detected vehicle. In some other situations particularly when there are bunch of vehicles near each other shadow of one vehicle may partially or completely be on another vehicle and this results in misdetection of these two or a group of separate vehicles as one big vehicle. Problems associated with occlusion would be created afterwards. As a result the performance of the surveillance system would be affected if the cast shadows are not detected and removed.

One of the applications of the shadow removal algorithms in traffic surveillance system is controlling the traffic of vehicles by employing a License Plate Recognition (*LPR*) system. An Intelligent Transportation System equipped with *LPR* has many applications such as flexible and automatic highway toll collection systems, analysis of city traffic during peak periods, enhanced vehicle theft prevention, effective law enforcement, highest efficiency for border control systems, building a comprehensive database of traffic movement, automation and simplicity of airport and harbor logistics, security monitoring of roads and etc.

In general the *LPR* system consists of three major parts [30]: License Plate detection, Character Segmentation and Character Recognition. A desired *LPR* system has to work under different imaging conditions such as low contrast, blurring, illumination and viewpoint changes. It is also supposed to perform properly in complex scenes and bad weather conditions. In addition response time is another restriction in real time applications such as license plate tracking. However, most of the license plate recognition algorithms work under restricted conditions, such as fixed illumination, limited vehicle speed, and stationary backgrounds.

1.1 Motivation

Management of present-day traffic in cities has become more important with the gradual increase in traffic flow and traffic violations. In cities with heavy traffic drivers tend to have a tendency to violate red lights and this behavior at times could lead to accidents and even deaths. To deter the violators many state of the art surveillance systems are being employed all over the world. The main aim of research carried out in this thesis was to develop the essential blocks of a system that could detect and identify red light violators in the city by the analysis of surveillance video taken from a fixed video camera. In order to speed up license plate processing and increase the accuracy of license plate detection it was decided that first the foreground containing the red light violator(s) would be separated from the background in the scene. To separate the foreground from the background in the scene a background subtraction algorithm has to be implemented. In this study a recently proposed state of the art *BG* modeling technique known as Group Based Histogram (*GBH*) algorithm has been adopted.

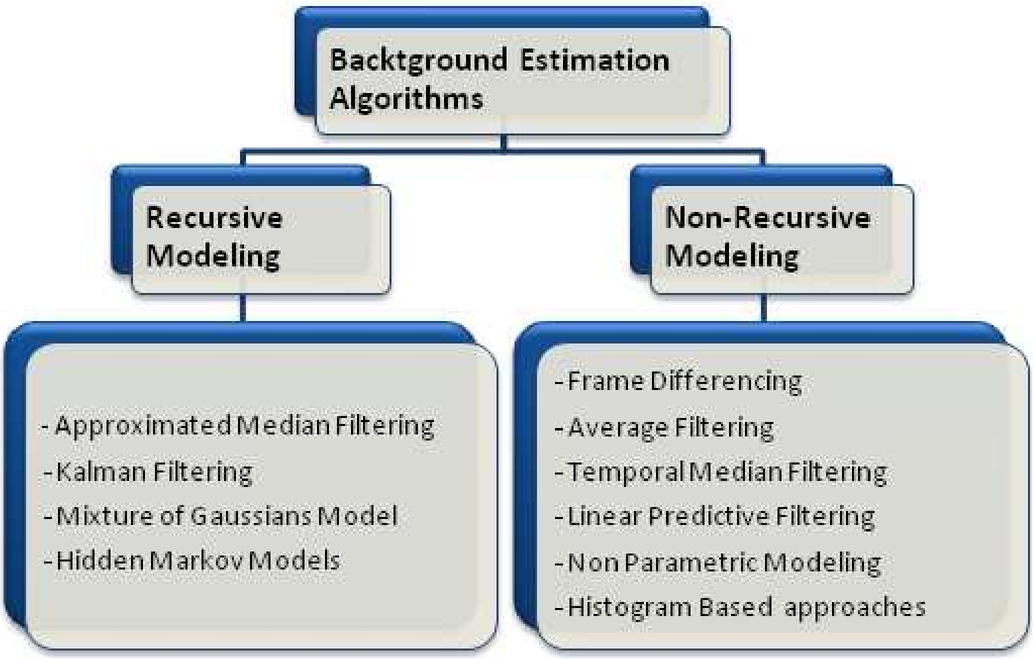
1.2 Related Works

First step of this research is background estimation / subtraction. Background estimation can be divided into two main categories: the predictive methods and the non-predictive methods. The predictive methods arrange the sequence as a time series and create a dynamical model at each pixel by considering the input frame, past observations and magnitude of difference between the actual observation and the predicted value. Non-predictive methods on the other hand, ignore the order of the input observations and develop a probabilistic model for each pixel. In [40] Elhabian states that background estimation algorithms can further be classified as non-recursive and recursive models.

Alternatively, background estimation methods can be categorized in to recursive modeling and non-recursive modeling techniques. A non-recursive technique estimates the background based on a sliding-window approach and recursive techniques update the background model using either a single or multiple component (distribution) models at each pixel of the frame observed. Oliver et al.

[41] used an Eigen-background subtraction method which adaptively builds an Eigen space that models the background method. A list of some non-recursive and some recursive background modeling techniques has been given in Table 1.1. Non-recursive modeling algorithms cover: frame differencing method in [57],[58], average filtering approach [60], median filtering [61],[9], minimum–maximum filtering method [62]. Recursive techniques include the approximated median filtering method [63], single Gaussian technique [64], Kalman filtering method [65], and Hidden Markov Models [66].

Table 1. 1 : Background Estimation Models



Over the past decades, several cast shadow detection methods have been introduced and classified in region-based and pixel-based groups or as model-base and shadow properties-base groups.

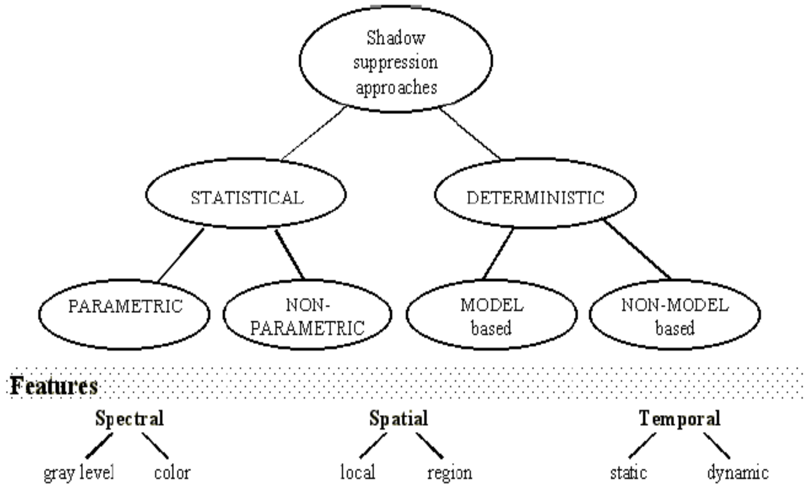
Many of these shadow detection algorithms are proposed for traffic surveillance. It has been demonstrated in [5] that shadows can be extracted by performing the difference between the current frame s_k (at time k) and a reference image s_0 that can be the previous frame, as in [6], or a reference frame, typically named "*background model*" as in [7][8][9].

Normally shadow detection algorithms are associated with techniques for moving objects segmentation. Some of these techniques are based on the inter-frame differencing [10],[11], background subtraction [12],[13], optical flow [14], statistical point classification [15],[16] or feature matching and tracking [17],[18].

There are two important shadow and object visual features that cause difficulties during shadow detection and removal. First, shadow points are detectable as foreground points as they differ significantly from the background. Second, shadow points have the same motion as the objects casting them. The goal of all proposed algorithms is to prevent moving shadows being classified as moving objects or parts of them, thus avoiding the merging of two or more objects into one and improving the accuracy and performance of object localization.

The approaches in literature differ by means of how they distinguish between foreground and shadow points. Most of these works locally exploit pixel appearance change due to cast shadows [8],[4],[16],[6]. A possible approach is to compute the ratio between the appearance of the pixel in the actual frame and the appearance in the reference frame as in [6]. Most of the proposed shadow removal algorithms take into account the model reported in [5], assume that camera and background are static and light source is strong enough. To give explanation for their differences, as demonstrated in Figure 1.2, four-class category of shadow detection algorithms are presented according to the decision process: Statistical Non-Parametric (*SNP*), Statistical Parametric (*SP*), Deterministic Model-based (*DM*) and Deterministic Non-Model-based (*DNM*).

Table 1. 2 : Shadow Detector Approaches Taxonomy



Generally speaking shadow regions are detected and removed based on the cast shadow observations of luminance, chrominance, and gradient density considering geometry properties in YC_bC_r color space domain. A combined probability map, called Shadow Confidence Score (*SCS*), of the region belonging to the shadow is deduced and using the computed scores shadow region are separated.

The deterministic class [4],[6],[13] can be further subdivided. Sub-classification can be based on whether the on/off decision can be supported by model based knowledge or not. Choosing a *model based* approach as in [20], [6] achieves undoubtedly the best results, but in most of the times, too complex and time consuming compared to the *non-model based* [9][23]. Moreover, the number and the complexity of the models increase rapidly if the aim is to deal with complex and cluttered environments with different lighting conditions, object classes and perspective views. It is also important to recognize the types of "features" utilized for shadow detection. Basically, these features are extracted from three domains: *spectral*, *spatial* and *temporal*. Approaches can exploit differently spectral features, i.e. using gray level or color information. Some approaches improve results by using spatial information working at a region level, instead of pixel level. Finally, some methods exploit temporal redundancy to integrate and improve results.

In the statistical methods as in [15],[26],[67] the parameter selection is a critical issue. Thus, we further divide the statistical approaches in *parametric methods* such as [15],[22],[23],[27] and *non-parametric* methods. In Parametric approach as in [15] an algorithm for segmentation of traffic scenes that distinguishes moving objects from their moving cast shadows has been proposed. A fading memory estimator calculates mean and variance of all three-color components for each background pixel. Given the statistics for a background pixel, simple rules for calculating its statistics when covered by a shadow are used. Then, MAP classification decisions are made for each pixel.

Furthermore, Xu et al. [22] assumed that shadow often appears around the foreground object and tried to detect shadows by extracting moving edges. Morphological filters were used intensively. Toth et al. [23] proposed a shadow detection algorithm based on color and shading information. This method changes the color space from *RGB* space to *LUV* space.

A contour based method for cast vehicle shadow segmentation in a sequence of traffic images taken from a stationary camera on top of a tall building is proposed by Yan et al. [27]. Xiao et al. [28] proposed a method of moving shadow detection based on edge information. Salvador et al. [29] introduced another method of shadow removal based on the use of invariant color models to identify and to classify shadows in digital images.

In literature, different approaches for applying license plate locating and recognition have been proposed. The features that license plate locating employed include shape, symmetry [43], height to width ration [44],[45], color [46],[45] texture of grayness [47],[45], special frequency [31] and variance of intensity values [49],[50].

License plate candidates determined by the plate localization stage are examined to be involved in character separation and character recognition stages. Different techniques used for character segmentation are projection [51],[52] morphology [47],[48],[50] connected components [45] and blob coloring. Every technique has its own disadvantages and advantages. The projection method assumes the orientation of a license plate is known and the morphology method requires the size of characters. In this research connected component technique is considered for character separation since numbers and characters in English are composed of one connected component region.

There is a large number of character recognition techniques reported. Some of them are based on Neural Networks [31], [32], [34],[35], Generic Algorithm [33], Edge Analysis [36],[37], Morphological Reconstruction [53], Markov Processes[38] and Invariants Moment calculations[39].

1.3 Thesis Structure

The thesis is organized in the following manner: first chapter includes introduction and a review on previous works. Chapter 2 introduces a Group-Based Histogram algorithm as a background estimation / subtraction method to segment moving foreground objects. In Chapter 3, three different shadow removal algorithms are discussed and evaluated. An application based on foreground/background separation, shadow detection and removal and license plate recognition processing is introduced in Chapter 4. Finally, Chapter 5 provides conclusion and future work.

CHAPTER 2

BACKGROUND ESTIMATION

2.1 Introduction

Group-Based Histogram (*GBH*) technique is a recently suggested method to generate a background model of each pixel from traffic image sequences. This algorithm features improved robustness against transient stops of foreground objects and sensed noise. Moreover, the method features low computational load, thus meets the real-time requirements in many practical applications. The proposed method has been used with vision-based traffic parameter estimation systems to segment moving vehicles from image sequences.

2.2 Group-Based Histogram

The *GBH* algorithm constructs background model using the histogram of intensities obtained from the current input frame and future frames at a specific location (x,y). Unlike other histogram based methods, group based histogram is forced to follow a Gaussian shaped trend to improve the quality of the estimated background [19].

Although the histogram approach is robust to the transient stops of moving foreground objects, the estimation is still less accurate than Gaussian Mixture Model

(*GMM*) in the case of non-static backgrounds (i.e. swaying grass, shaking leaves, rain etc.). The proposed *GBH* method effectively exploits an average filter to smoothen the frequency curve of '*conventional*' histogram. From a smoothed histogram a more accurate mean value and respectively standard deviation can be estimated. One can easily and efficiently estimate the single Gaussian model constructed by background intensities from image sequences during a fixed span of time.

While doing background estimation based on histogram analysis, the intensity with the maximum frequency in the histogram is treated as background, because each intensity frequency in the histogram is proportional to its occurrence probability. The background intensity can therefore be determined by analyzing the intensity histogram. However, sensing variation and noise from image acquisition devices or pixels having complex distributions may result in erroneous estimates. This may cause a foreground object to have the maximum intensity frequency in the histogram.

Since the maximum frequency of the histogram indicates the intensity of the pixel belonging to the background model, there will not be any inclusion of slow moving objects or transient stops in the detected foreground. However, the maximum peak of the conventional histogram of each pixel will not necessarily locate the intensity of background model at that specific pixel. In some cases this maximum may not be unique so further processing may be needed to compensate the loss which will affect the real time tracking.

In group based histogram each of the individual intensities is considered along with its neighboring intensity levels and forms an accumulative frequency. The frequency of coming intensity is summed up with its neighboring frequency to create a Gaussian shape histogram.

The accumulation can be done by using an average filter of width $2w+1$ where w stands for half width of the window. The output $n^*_{u,v}(l)$ of the average filter at level l can be expressed as:

$$n^*_{u,v}(l) = \sum_{r=-w}^{w} n_{u,v}(l+r); \qquad 0 \le l+r \le (L-1) \tag{2.1}$$

where $n_{u,v}(l+r)$ is the count of the pixel having the intensity $l+r$ at the location (u, v) and L is the number of intensity levels based on the number of bits in each layer.

Maximum probability density $p^*_{u,v}$ of a pixel at location (u, v) over the recorded image frames can be computed through a simple division of the occurrence for a pixel by,N^*, the total frequency of the *GBH*.

$$p^*_{u,v} = \frac{max_{0 \le l \le L-1}\{n^*_{u,v}(l)\}}{N^*} \tag{2.2}$$

If the width of the window is chosen to be less than a preset value, the location of the maximum will be closer to the center of the Gaussian model than the normal histograms. This is the result of the smoothening effect by the filter used. Therefore the mean intensity of the background model will be:

$$\mu_{u,v} = \arg max_l\{n^*_{u,v}(l)\} \tag{2.3}$$

For smaller window widths, the computational time will be less and the accuracy of the background pixels estimates will vary for different window sizes and standard deviations. To show how window width can be selected an example based on 13 Gaussians generated by Gaussian random number generator can be given. The mean for each Gaussian has been chosen as 205 and standard deviations varied between 3 and 15. The percentage of errors while trying to estimate the background pixels using the conventional histogram approach versus the *GBH* method are depicted in Table 2.1. The window widths and the range of standard deviate values (3-15) used in the comparisons have also been shown in the table.

Table 2. 1 : Error Estimation for the Gaussian mean using conventional histogram and GBH methods

Standard deviation	3	4	5	6	7	8	9	10	11	12	13	14	15
Hist.	−1.5%	−1.5%	**−2.0%**	**−2.4%**	**−2.4%**	**−2.9%**	**−3.4%**	**−2.9%**	**−2.9%**	**−4.4%**	**−2.9%**	**−4.9%**	**−4.4%**
Width w	Estimation result of the proposed GBH												
1	0.5%	1.0%	0.0%	−0.5%	−0.5%	**2.0%**	**−3.4%**	**−2.4%**	**−2.9%**	−1.5%	**−3.4%**	0.5%	1.0%
2	0.5%	0.0%	0.0%	−1.5%	−1.0%	1.0%	**−2.4%**	**−2.0%**	**−2.4%**	**−2.0%**	**−3.9%**	1.0%	1.5%
3	0.0%	0.0%	−0.5%	−1.0%	0.5%	−1.5%	**−2.4%**	**2.0%**	**2.0%**	**−2.4%**	−1.5%	1.5%	**-2.9%**
4	0.5%	0.5%	0.0%	−0.5%	−0.5%	0.5%	−1.5%	−1.0%	−1.5%	**−2.4%**	−1.0%	**2.4%**	**-2.4%**
5	0.5%	0.5%	−0.5%	0.0%	0.0%	−0.5%	−1.0%	−0.5%	−0.5%	**−2.0%**	−1.5%	0.5%	0.5%
6	0.5%	0.0%	−0.5%	−0.5%	−0.5%	−1.0%	−0.5%	0.0%	−0.5%	−1.5%	−1.5%	0.0%	−1.5%
7	0.5%	0.5%	−0.5%	0.5%	0.0%	−0.5%	0.0%	0.0%	0.0%	−0.5%	−1.5%	1.0%	−1.0%

The results prove the superiority of implementing *GBH* method to conventional histograms. Considering the simulation results it can be concluded that a greater window width will be needed for high-accuracy performance as the standard deviation increases. According to the simulation results and error rate of mean estimation within ± 2 %, the width, w, can be determined as follows [19]:

$$w = \begin{cases} 3 & 3 \leq \sigma_i \leq 7 \\ 5 & 8 \leq \sigma_i \leq 10 \\ 7 & 10 \leq \sigma_i \leq 15 \end{cases} \tag{2.4}$$

Where, σ_i represents the standard deviation of the original Gaussian.

As mentioned before, the mean intensity $\mu_{u,v}$ can be computed by selecting the maximum frequency of the smoothened histogram. When a new intensity l is captured, the algorithm does not process all the possible intensities because there would be very few neighboring pixels that fall in the selected window together with the input intensity.

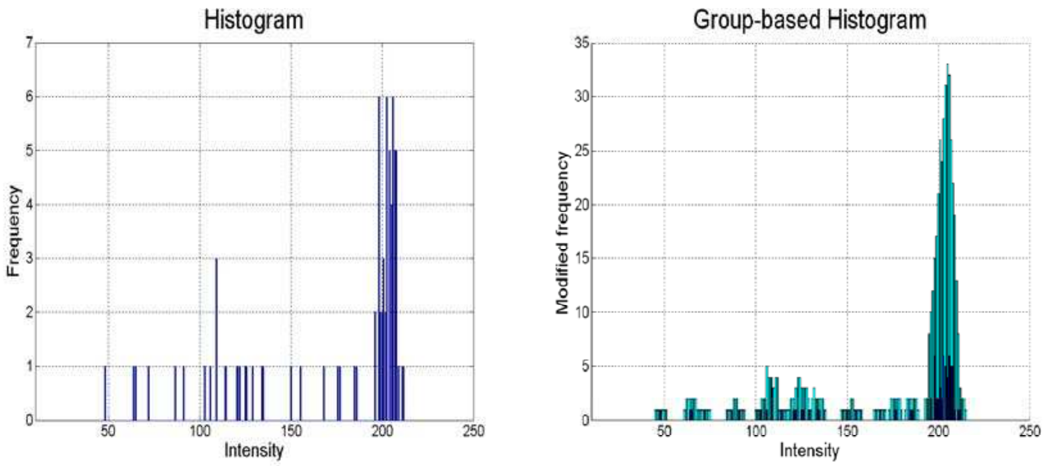
The steps of the procedure for estimating the mean of the distribution are as follows. First, the current intensity l of the pixel is recorded. Second, the occurrence frequency of that intensity and the intensities of neighbors from l-w to l+w is incremented by one. Finally, the new maximum value is checked to see if it is greater than the previously estimated mean value or not. If the condition is satisfied then replacement of the former mean with the new one is made and then the algorithm will return to the first step.

After computing the mean intensity of the Gaussian shaped histogram, the variance could also be estimated using the following expression:

$$\sigma_{u,v}^2 = \frac{1}{\sum_{x=\mu_{u,v}-3\sigma'}^{x=\mu_{u,v}+3\sigma'} n_{u,v}(x)} \times \sum_{x=\mu_{u,v}-3\sigma'}^{x=\mu_{u,v}+3\sigma'} (x-\mu_{u,v})^2 n_{u,v}(x) \qquad (2.5)$$

where, σ' is the maximum standard deviation of the Gaussian.

Figure2.1 (b) demonstrates the histogram smoothing after the implementation of average filtering window for a certain pixel in a traffic video sequence. From Figure 2.1 (a) one can conclude that it would be possible to model the results with a Gaussian distribution. However since several peaks with similar frequencies occur in the histogram, selecting the mean is not straight forward. By applying the windowing technique proposed in *GBH* the histogram will be smoothed and this multiple peaks are eliminated.



(a) (b)

Figure 2. 1: Statistic analysis of pixel intensity

(a) Histogram (b) group-based histogram

To cope with the illumination changes of the environment, the histogram can be re-built after every 15 minutes.

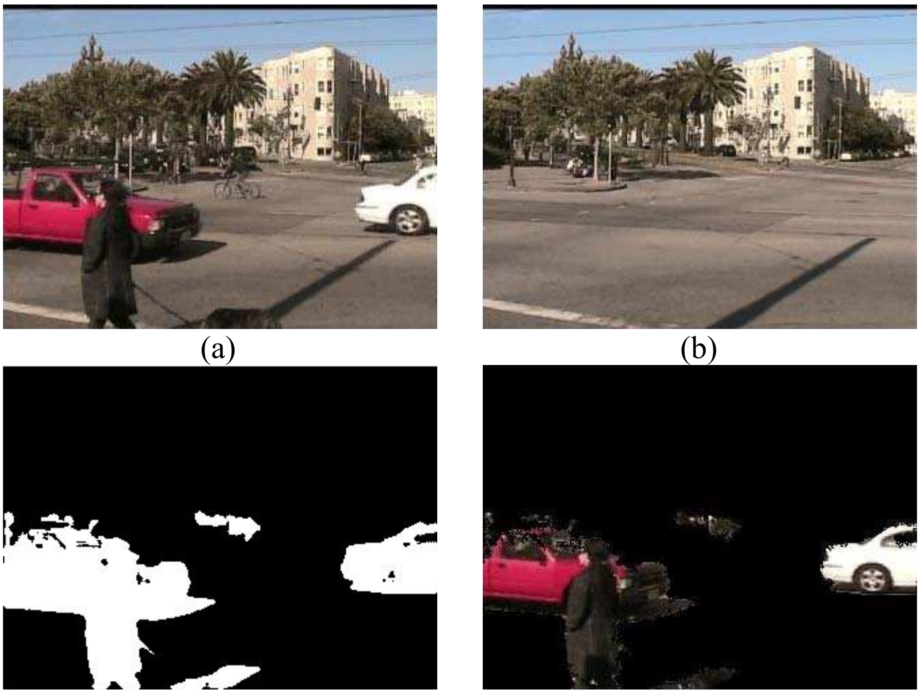
2.3 Foreground Segmentation

If the current pixel intensity under observation is to be accepted as foreground, its distance from the mean of the distribution should not exceed three times the standard deviation of the distribution. With this restriction a pixel at location (u,v) on the image, can be assumed as a part of the foreground mask as shown by equation 2.6 below:

$$F(u,v) = \begin{cases} 1, & (moving\ objects) \quad if\,|I(u,v) - \mu(u,v) > 3\sigma(u,v)| \\ 0, & otherwise \end{cases} \tag{2.6}$$

where, $\mu(u,v)$,$\sigma(u,v)$ represent mean and standard deviation of the background model at location (u,v).

Figure 2.2 provides an example for background estimation by applying the *GBH* approach on a video sequence at a junction. The segmented foreground objects are vehicles and pedestrians with their corresponding cast shadows. On the segmented foreground objects shadow removal algorithms are applied in order to get vehicles without cast shadows.



(a) (b)

(c) (d)

Figure 2. 2: Foreground estimation using GBH technique

(a) An Input Frame of the Sequence (b) Estimated Background

(c) Moving Foreground Image Mask (d) Extracted Foreground

CHAPTER 3

SHADOW REMOVAL ALGORITHMS

As mentioned earlier, when the detected foreground mask contains shadows, the calculated quantities such as location, dimension, speed, and number of vehicles often include large errors. For instance, in a traffic scene with detached shadows approximately the same size as the car, a vehicle's location may be incorrectly estimated as the shadow region. Long shadows could also connect two separate vehicles as if they were a single object. Therefore, the performance of the overall system may be seriously affected if the cast shadow is not detected and removed efficiently. Below three different algorithms are introduced and compared against each other for efficient and reliable detection of cast shadows.

3.1 Shadow Confidence Score Based Shadow Detection

3.1.1 Introduction

The robust method described in [54] has adopted YC_bC_r color space for detecting cast shadows of moving vehicle in a monocular color traffic image sequence. Firstly the background estimation/subtraction algorithm is used to generate the foreground mask and then extracted blobs corresponding to binary mask locations in the color image are converted to YC_bC_r.

The extracted foreground mask generally includes both the moving vehicles and their cast shadows as a binary map. In [54] the mask is referred to as the Moving Foreground Mask (*MFM*). From this *MFM* the Shadow Confidence Score (*SCS*) can be calculated to indicate the likelihood of shadow according to the cast shadow characteristics. The edge pixels of the input image within the *MFM* are classified into object-edge pixels and non-object edge pixels. Then object-edge pixels are bounded by a convex hull to generate a more accurate foreground mask of the moving vehicles.

3.1.2 Methodology

3.1.2.1 *RGB* to YC_bC_r Conversion

YC_bC_r is an encoded nonlinear *RGB* signal. The *Y*-component is known as the luminance value and is a weighted sum of the *R*, *G*, *B* components. "C_r" and "C_b" are formed by subtracting the luminance component from red and blue components respectively and multiplying the results by some weight factor. In this work the YC_bC_r color space was chosen since it can separate luminance from color components. This was a good idea since luminance values for shadow regions and non-shadow regions would significantly vary from each other. The command used for converting the color *RGB* image to YC_bC_r was RGB2YCBCR(.).

The Figure 3.1 and the equations given below show how one can transform an *RGB* image into the YC_bC_r domain.

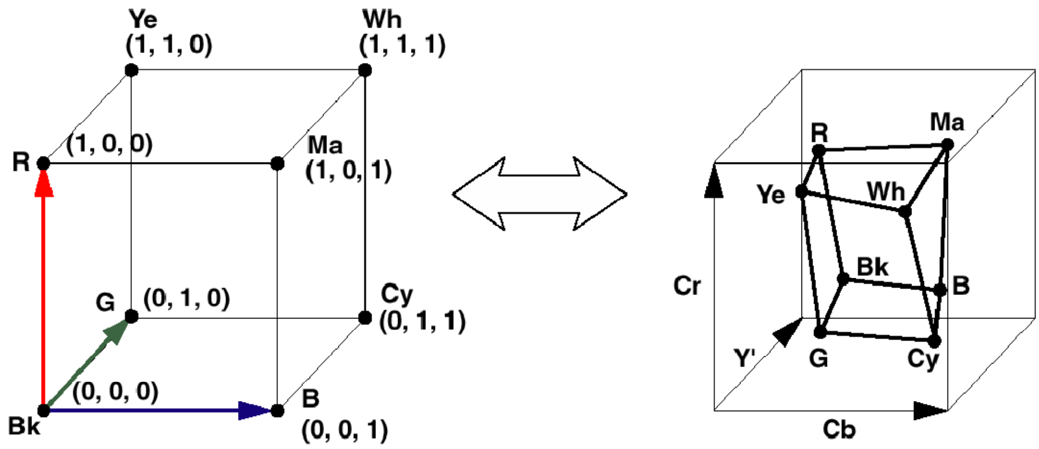


Figure 3. 1: *RGB* to YC_bC_r Conversion

(3.1)

$$C_r = 0.7132\ (R - Y) \tag{3.2}$$

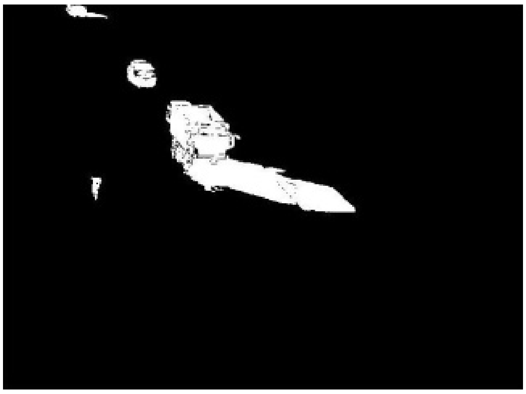
$$C_b = 0.5647\ (B - Y) \tag{3.3}$$

3.1.2.2 Observations about cast shadows

In essence shadows can be classified as: the self -shadow and the cast shadow. Self shadow is the part of the object that is not illuminated by direct light, while cast shadow is the region projected by the object in the direction of direct light. Even though changes in illumination and weather conditions could lead to cast shadows that have different colors or tones, [54] states four generic features that are generally true about cast shadow. In the sections that follow we try to explain these four observations with the help of some examples. For identifying the correct shadow pixels the SCS based processing would require the input frame, the estimated background and the foreground binary mask as depicted in Figure 3.2 (a) –(c).

(a) (b)



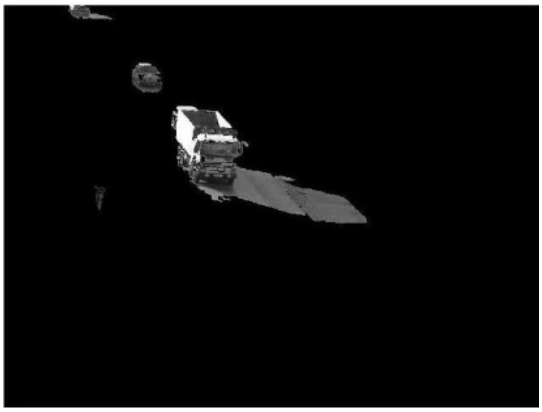
(c)

Figure 3. 2: An outdoor background estimation and foreground segmentation (a) Input frame, (b) Estimated Background, (c) Moving Foreground Mask

Figure 3.2 (a) shows an input frame containing a truck in an outdoor traffic scene under bright sunlight with corresponding cast shadow. Figure 3.2 (b) represent estimated background and Figure 3.2 (c) is the Moving Foreground Mask (*MFM*) gained from difference of input frame and estimated background (a morphological closing has been applied to join the discontinuities of the object). The small holes in the foreground mask image are due to the similarities between the vehicle colors and corresponding background that is subtracted from the input frame.

Observation1. *The Luminance values of the cast shadow pixels in the input are lower than those of the corresponding pixels in the background image.*

As stated in [54], the cast shadow region is the darker region due to its lower luminance values. The Figure 3.3 (a) demonstrates the luminance of the truck within the mask region in the input frame and the luminance of the corresponding background within the mask is depicted in Figure 3.3 (b). The subtraction between the two masked regions is shown in Figure 3.3 (c). It's obvious from the figures that luminance of input image is most of the time lower than the background image in the cast shadow region.



(a)

(b)

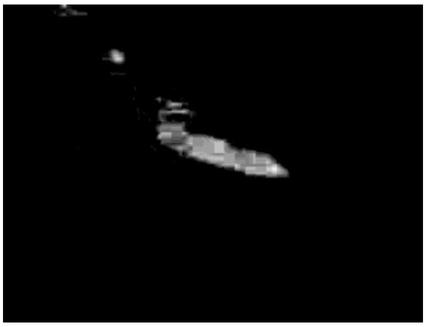
(c)

Figure 3. 3: Luminance of masked input image and of the corresponding background

(a) Luminance of masked input frame, (b) Luminance of masked background frame (c) Luminance difference between masked input and background frames

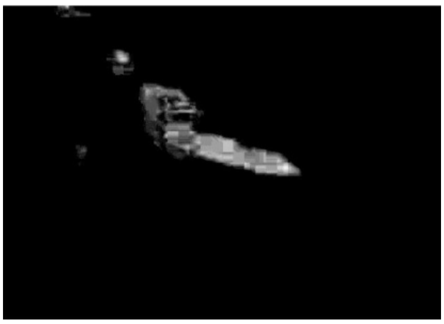
Observation2. *The chrominance values of the cast shadow pixels are identical or only slightly different from those of the corresponding pixels in the background image.*

The chrominance feature of foreground vehicle with its cast shadow based on observation 2 is depicted in Figure 3.4. Luminance and chrominance components of the images are separated in YC_bC_r color space. C_r and C_b components of masked input frame and masked background frame are calculated separately. Absolute difference between C_b component of masked input frame and background frame are then taken (Figure 3.4 (a)). Similarly, the absolute difference of the C_r components is depicted by Figure 3.4 (b). Finally, the sum of C_b and C_r absolute differences is calculated and shown in Figure 3.4 (c). For typical sunlight, a decrease in illumination will cause only a slight change in chrominance values of the shadow pixels in both the masked input and the masked estimated background images.



(a)

(b)



(c)

Figure 3. 4: Chrominance of masked input frame and of the corresponding background image
(a) C_b differences $|C_{bI} - C_{bB}|$, (b) C_r differences $|C_{rI} - C_{rB}|$, (c) Chrominance difference $|C_{bI} - C_{bB}| + |C_{rI} - C_{rB}|$

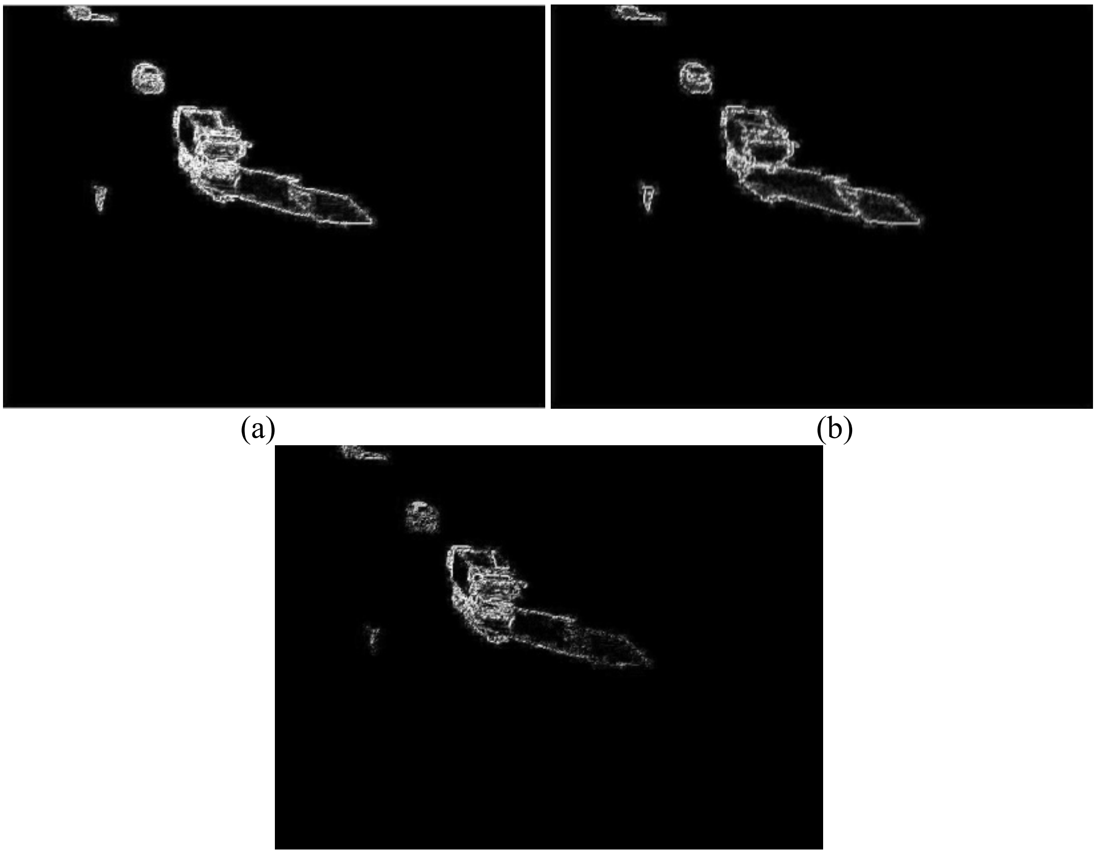
Observation3. *The difference in gradient density values of the cast shadow pixels and the corresponding background pixels is relatively low. The difference in gradient density values between the vehicle and the corresponding background pixels is relatively high.*

Gradient Density (*GD*) is the average of gradient magnitudes over a local area which can be computed using a spatial window as shown in the equation below:

$$GD(x,y) = \frac{1}{(2\omega+1)^2} \sum_{i=x-\omega}^{x+\omega} \sum_{j=y-\omega}^{y+\omega} |G_x(i,j) + G_y(i,j)| \tag{3.4}$$

Where, $G_x(i,j), G_y(i,j)$ are horizontal and vertical edge magnitude obtained using 'Laplacian' gradient operator for pixel *(i,j)* and *(2ω +1)* is the spatial window size.

According to Figure 3.5 (c) there is no significant gradient density difference in the cast shadow region, but in the vehicle region the gradient density difference between the masked input and background images varies considerably. Therefore, one can assume that the majority of vehicle region pixels have large gradient density differences.



(a) (b) (c)

Figure 3. 5: Gradient Density of masked input frame and of the corresponding background image

(a) gradient density of the mask input, (b) gradient density of masked background frame, (c) gradient density difference

Observation4. *The vehicle can be bounded approximately by means of a convex mask. The cast shadow is an extension of the object mask.*

The cast shadow can be separated from the foreground object based on the shadow confidence scores and the object edge pixels of the foreground masked input image. First, all the pixels with significant gradient values are detected using the edge detector within the *MFM*. Then from the selected pixels the ones with high shadow confidence scores are discarded using a threshold value. Finally a convex hull is fitted to the remaining pixels to generate a binary mask for the detected foreground object. Figure 3.6 provides an example of this processing.

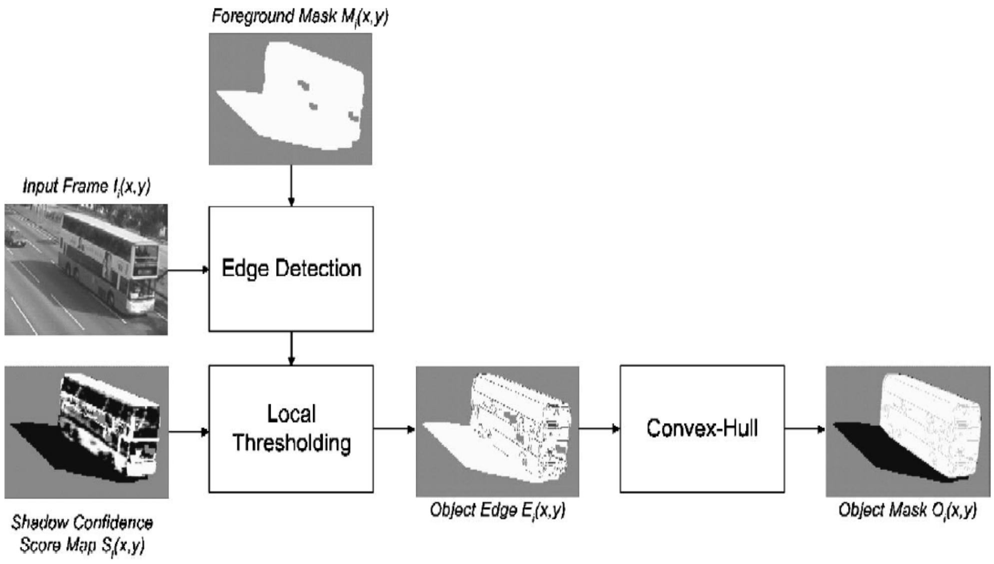


Figure 3. 6: Object and cast shadow separation using a convex hull [54]

3.1.3 SCS Calculation

As explained in [54], [68] Luminance, chrominance and gradient density of each pixel are to be calculated from the input and background images in the region shown by *MFM*. Calculation of overall score $S_i(x,y)$ needs three mapping functions to be defined: Luminance Score $S_{L,i}(x,y)$, Chrominance Score $S_{c,i}(x,y)$, Gradient Density Score $S_{G,i}(x,y)$.

3.1.3.1 Luminance Score

In [68] luminance score is defined by means of the luminance difference and the related luminance score. They can be computed based on the expressions given by equations 3.5 and 3.6. $L_i(x,y)$ is the luminance difference between the i^{th} input image and the i^{th} background image at location (x,y) where the *MFM* value is 1.

$$L_i(x,y) = l_{I,i}(x,y) - l_{B,i}(x,y) \tag{3.5}$$

$$S_{L,i}(x,y) = \begin{cases} 1 & L_i(x,y) \le 0 \\ \dfrac{[T_L - L_i(x,y)]}{T_L} & 0 < L_i(x,y) < T_L \\ 0 & L_i(x,y) \ge T_L \end{cases} \tag{3.6}$$

T_L is a predefined threshold to accommodate the acquisition noise in luminance domain. $l_{i,I}(x,y)$ and $l_{i,B}(x,y)$ are the luminance of the input frame and background at pixel location *(x,y)*. The initial value of the threshold T_L has been taken from ref. [54], then to improve shadow detection results for the custom videos best values for thresholds has been selected experimentally.

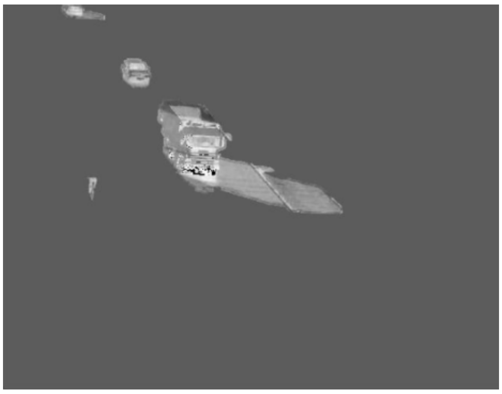


Figure 3. 7: Luminance Score of the masked input frame

Also since the luminance values for pixels in the masked input image are lower than that of the corresponding pixels in the masked background image for cast shadow regions a negative luminance difference value would indicate that the pixel of interest belongs to the cast shadow region.

3.1.3.2 Chrominance Score

According to the information given in [68], the chrominance difference and the chrominance score related to this difference can be computed using equations 3.7 and 3.8. $C_i(x,y)$ is the chrominance difference between the i^{th} input image and the i^{th} background image at location *(x,y)* where the *MFM* value is 1.

$$C_i(x,y) = \left|Cb_{I,i}(x,y) - Cb_{B,i}(x,y)\right| + \left|Cr_{I,i}(x,y) - Cr_{B,i}(x,y)\right| \quad (3.7)$$

$$S_{C,i}(x,y) = \begin{cases} 1 & C_i(x,y) \le T_{C1} \\ \dfrac{[T_{C2} - C_i(x,y)]}{T_{C2} - T_{C1}} & T_{C1} < C_i(x,y) < T_{C2} \\ 0 & C_i(x,y) \ge T_{C2} \end{cases} \quad (3.8)$$

T_{C1} and T_{C2} are predefined thresholds to accommodate the tolerance to acquisition noise in the chrominance domain. $C_{b,i,I}$, $C_{b,i,B}$, $C_{r,i,I}$ and $C_{r,i,B}$ are the chrominance values of the input frame and background at pixel *(x,y)*. Similar to the luminance threshold, initial values for T_{C1} and T_{C2} have been selected from [54] and then threshold values have been optimized by trial and error approach.

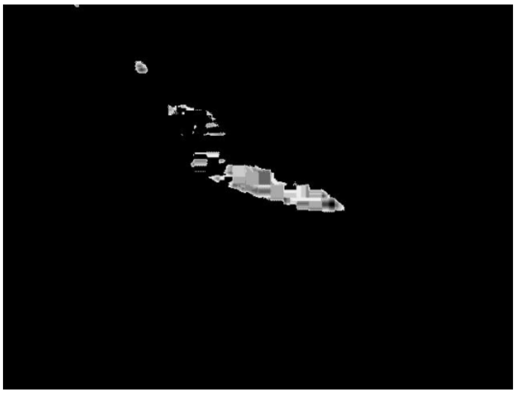


Figure 3. 8: Chrominance score of the masked input frame

As stated in Observation-2 the chrominance value of a pixel in the masked input image is approximately the same as that of the corresponding pixel in the masked background image at the cast shadow region.

3.1.3.3 Gradient Density Score

The gradient density difference and the related gradient score can be computed using equations 3.9 and 3.10. $GD_i(x,y)$ is the gradient density difference between the i^{th} input image and the i^{th} background image at location *(x,y)* where the MFM value is 1.[68]

$$GD_i(x,y) = GD_{I,i}(x,y) - GD_{B,i}(x,y) \tag{3.9}$$

$$S_{G,i}(x,y) = \begin{cases} 1 & GD_i(x,y) \leq T_{G1} \\ \dfrac{[T_{G2} - GD_i(x,y)]}{T_{G2} - T_{G1}} & T_{G1} < GD_i(x,y) < T_{G2} \\ 0 & GD_i(x,y) \geq T_{G2} \end{cases} \tag{3.10}$$

Here, T_{G1} and T_{G2} are two predefined thresholds and $GD_{i,I}(x,y)$ and $GD_{i,B}(x,y)$ are the average of the gradient magnitudes over a spatial window area in the masked input frame and corresponding masked background at pixel *(x,y)*.

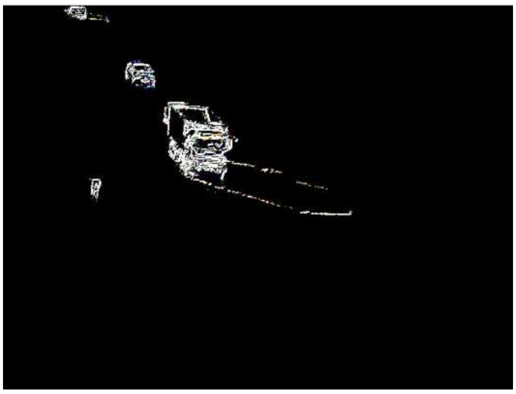


Figure 3. 9: Gradient Density Score

According to Observation-3, the gradient density values are mostly canceled out in the cast shadow region and a pixel with small gradient density difference value is more likely to be a part of the cast shadow region.

3.1.3.4 Combined SCS

By combining the three calculated scores, $S_{L,i}(x,y)$, $S_{C,i}(x,y)$ and $S_{G,i}(x,y)$, the total shadow confidence score $S_i(x,y)$ can be obtained as :

$$S_i\,(x,y) = \zeta\left[S_{L,i}(x,y), S_{C,i}(x,y), S_{G,i}(x,y)\right] \tag{3.11}$$

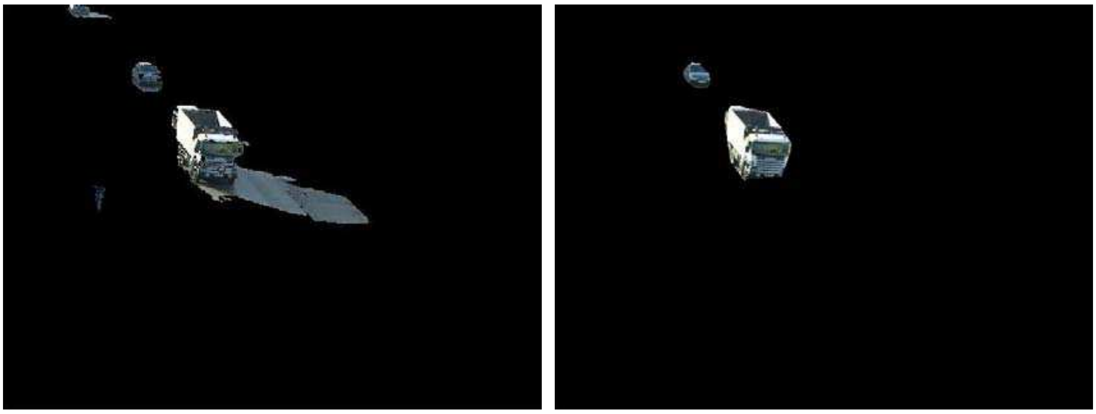
Where, "ζ" denotes the logical *AND* operation.

Figure 3. 10: Total Shadow Confidence Score (SCS)

A pictorial representation of the overall shadow confidence score is depicted in Figure 3.10.

3.1.4 Moving Cast Shadow Detection and Elimination

Moving cast shadow detection and removal is done in two stages. First, pixels with lower gradient density are removed using a *Canny* edge detector within the mask. Remaining pixels are denoted as *E1*. Second, since shadow pixels result in higher total shadow confidence score a threshold could be selected for filtering out the pixels with high SCS values. Pixels with corresponding SCS that is above the threshold T_s can be categorized as shadow and set to zero. The final outcome has most of the shadow pixels eliminated from the foreground mask. In order to crop out a foreground object with no defects (holes and noise) a convex hull can finally be applied to the remaining set of pixels and the object is selected using this new hull-based mask. Figure 3.11 (a) and (b) show the masked input frame with and without cast shadows.



(a) (b)

Figure 3. 11: SCS shadow removal Algorithm

(a) Masked input frame with shadow, (b) shadow removed frame

To prevent misdetection of two separate vehicles as one, it is necessary to take into account that the total number of pixels in the object mask can not exceed a pre-defined threshold value. In general, in order to assign a suitable threshold value during the detection procedure T_{width} and T_{lenght} are assumed to limit the size of detected vehicles. Since the width of a typical bus or truck is not wider than the road lane, thresholds are defined as T_{width} = *lane width* × *2/3* and T_{lenght} = *large vehicle length.*

3.2 Shadow Suppression in *HSV* Color Space

3.2.1 Introduction

Another method for shadow detection and removal is based on the *HSV* color space. The 'Statistical and Knowledge Based Object Tracker' (*SAKBOT*) system tries to suppress the shadows using the *HSV* color space [13]. *HSV* color space corresponds closely to human perception [2] of color and its mask is more accurate than *RGB* color space to detect shadow regions.

3.2.2 Methodology

3.2.2.1 RGB to HSV Conversion

HSI, *HSV*, *HSL* (Hue, Saturation, Intensity/Value/Lightness) are Hue-saturation based color spaces. They are ideal when developing image processing algorithms based on color descriptions that are natural and intuitive to human perception. Hue is a color attribute that describes a pure color (pure yellow, orange or red), whereas saturation gives a measure of the degree to which a pure color is diluted by white light. Intensity is a subject descriptor that is particularly impossible to measure. The intuitiveness of *HSV* color space components as an explicit discrimination between luminance and chrominance properties made these color spaces work desirably well on traffic surveillance and shadow removal algorithms.

The main reason for usage of *HSV* color space is that it explicitly separates chromacity and luminosity for assessing the effect of occlusion due to shadow changes on the *H*, *S* and *V* components.

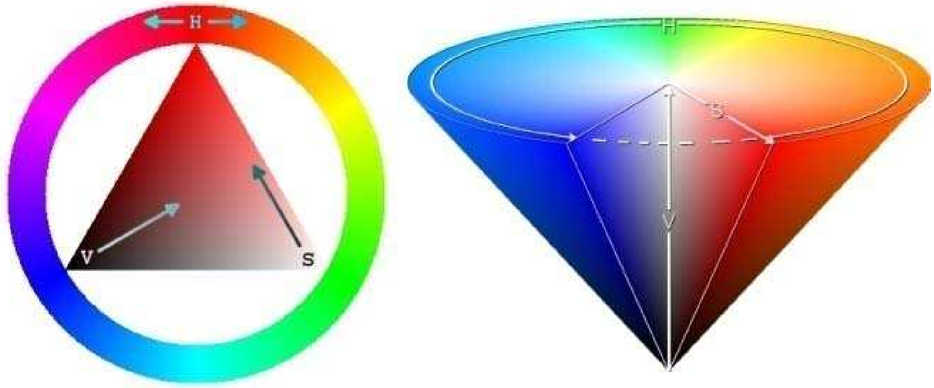


Figure 3. 12: Wheel and conical representation of *HSV* color model

RGB Space to *HSV* Space Transformation equations:

$$H = \begin{cases} \theta & if\ B \leq G \\ 360 - \theta & if\ B > G \end{cases} \tag{3.12}$$

With

$$\theta = \cos^{-1}\{\frac{\frac{1}{2}[(R-G)+(R-B)]}{[(R-G)+(R-B)(G-B)]^{1/2}}\} \tag{3.13}$$

The saturation component is given by

$$S = 1 - \frac{3}{(R+G+B)}[\min(R,G,B)] \tag{3.14}$$

And the intensity and value component is given by

$$V = \frac{1}{3}(R+G+B) \tag{3.15}$$

RGB values have been normalized to the range [0, 1] and the angle θ is measured with respect to the red axis of the *HSV* space.

3.2.2.2 Algorithm

According to *SAKBOT* [6] in order to analyze shadow region only pixels which are estimated as moving objects (vehicle plus the corresponding shadow) are to be considered. These pixels are detected with high differences according to equation3.16 and the ratio in equation 3.17.

$$D_k(x,y) = S_{k+1}(x,y) - S_k(x,y) \tag{3.16}$$

Where S_k is the luminance of the pixel in location *(x,y)* which is obtained from $S_k(x,y) = E_k(x,y) \times \rho_k(x,y)$ at time instant k. $E_k(x,y)$ is the irradiance and it is computed as:

$$E_k(x,y) = \begin{cases} C_A + C_p \cos(\boldsymbol{N}(x,y), \boldsymbol{L}) & illuminated \\ C_A & shadowed \end{cases} \tag{3.17}$$

Where C_A and C_P are the intensity of the ambient light and the light source respectively. L is the direction of the light source and $N(x,y)$ is the object surface normal. $\rho_k(x,y)$ is the reflectance of the object surface. Firstly it is assumed that the light source is strong, second camera and background are static which would result in static reflection $\rho(x,y)$ and third, background is planar.

Local appearance changes due to cast shadows can be computed by ratio $R_k(x,y)$ in equation 3.18.

$$R_k(x,y) = \frac{S_{k+1}(x,y)}{S_k(x,y)} \tag{3.18}$$

This ratio is less than one for shadow pixels. In fact cast shadow pixels darken the background image but vehicle pixels may or may not darken the background depending on the object color. Another interesting point is that shadows often lower the saturation of the pixels.

If in equation 3.18 $S_k(x,y)$ is approximated with intensity value (*V*-component) of the pixel in the *HSV* color space at location (x,y) in the time instant k, then a shadow point mask $SP_k(x,y)$ for each pixel can be defined as :

$$SP_k(x,y) = \begin{cases} 1 & if\ \alpha \le \dfrac{I_k^V(x,y)}{B_k^V(x,y)} \le \beta \\ & \wedge \left(I_k^S(x,y) - B_k^S(x,y)\right) \le \tau_S \\ & \wedge \left|I_k^H(x,y) - B_k^H(x,y)\right| \le \tau_H \\ 0 & otherwise \end{cases} \tag{3.19}$$

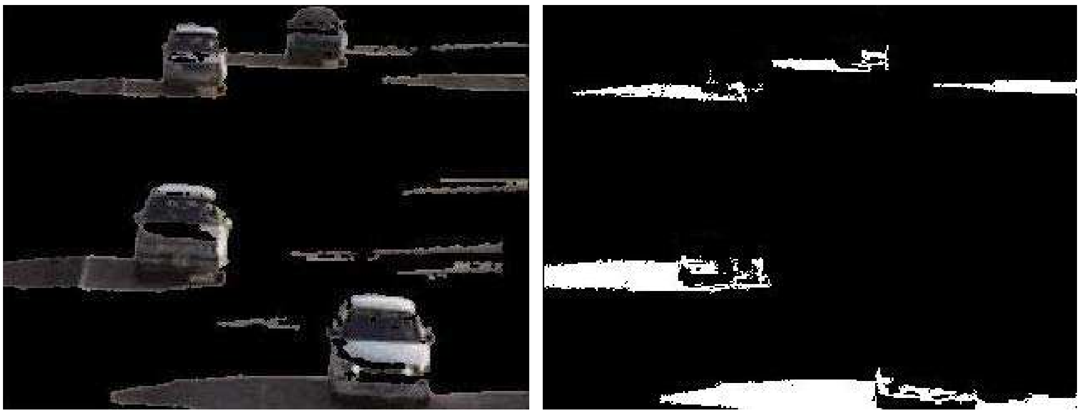
Where $I_k^S(x,y), I_k^H(x,y), I_k^V(x,y)$ are *HSV* components of the input frame at time instant k and location (x,y) and $B_k^S(x,y), B_k^H(x,y), B_k^V(x,y)$ are *HSV* components of the background frame at time instant k and location (x,y).

Figure 3.13 (a) below shows one selected frame from the Yeni-İzmir junction of Famagusta with its corresponding shadows and Figure 3.13 (b) depicts the HSV detected shadows for this frame. Similarly Figure 3.14 shows the foreground and segmented shadow regions for a selected frame of the Highway-I test sequence from *VISOR*.

(a) (b)

Figure 3. 13: Shadow mask of a video frame at a junction in Famagusta

(a) Foreground with its corresponding shadow, (b) Shadow Point Mask



(a) (b)

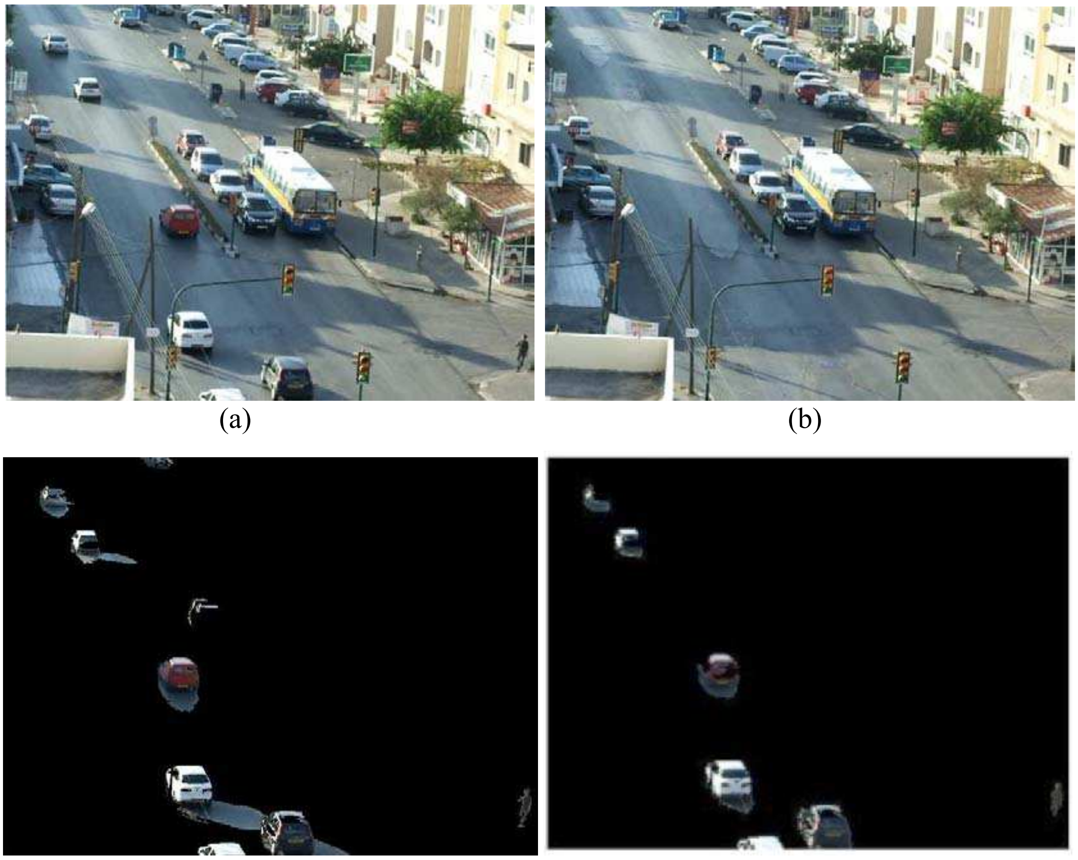
Figure 3. 14: Shadow mask of highway-I video [56]

(a) Foreground with its corresponding shadow, (b) Shadow Point Mask

In equation 3.19 the lower bound α is used to define a minimum value for the darkening effect of shadows on the background and it is almost proportional to the light source intensity and the upper bound β prevents the system from identifying noise which slightly changes the background in the shadow regions.

It has been shown that the chrominance values for both the shadow and non-shadow pixels would vary only slightly. The choice of τ_H and τ_S is done according to this assumption. This choice is complicated and the threshold values have to be chosen by trial and error.

As shown in Figure 3.15 (d) once the shadow pixels are detected and suppressed the new foreground would only contain the moving vehicles.



(a) (b)

(c) (d)

Figure 3. 15: *HSV* color space result on shadow removal purpose

(a) Input Frame, (b) Estimated Background, (c) Extracted Foreground (d) shadow removed foreground

3.3 Hybrid color and Texture Based Shadow Removal

In this statistical approach it is assumed that irradiation consists of only one light source and the chromaticity in the shadow region should be the same as when it is directly illuminated. A hybrid color and texture model is employed to assist distinguishing of shaded backgrounds from the ordinary background or of moving *FG* objects.

3.3.1 Color Based Analysis

The hybrid color technique proposed in [26] makes use of the RGB color space. In the RGB domain ambient light is ignored and RGB space is invariant to changes of surface orientation relatively to the light source.

On perfectly matte surfaces, the perceived color is the product of illumination and surface spectral reflectance [26]. Therefore, if a method could separate the brightness from the chromaticity component then the observation will be independent of illumination changes. Figure 3.16 illustrates the color model in three dimensional RGB space.

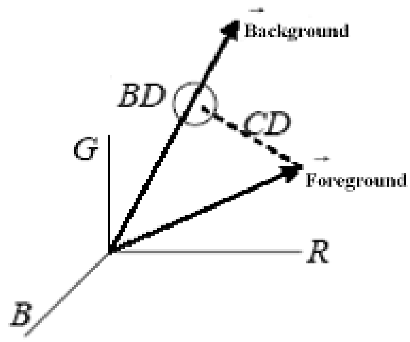


Figure 3. 16: Distortion measurements in the *RGB* Space

Here, foreground denotes the RGB value of a foreground pixel in the incoming frame and background is of its background counterpart.

To detect the shadow due to illumination on matte surface the distortion of input frame from background frame can be measured. This distortion is decomposed to *Brightness Distortion* and *Chromaticity Distortion.* [26].

3.3.1.1 Brightness Distortion

Brightness Distortion (BD) is a scalar value that brings the observed color close to the expected chromaticity line. It is obtained by minimizing

$$BD = (inputframe - \alpha\ estimated\ background\ frame)^{2} \qquad (3.20)$$

α represents the pixels brightness strength with respect to an expected value. α is "1" if the brightness of given pixel in the input frame is the same as in background frame. α is less than "1" if it is darker than background frame and it is greater than "1" if it becomes brighter than expected brightness.

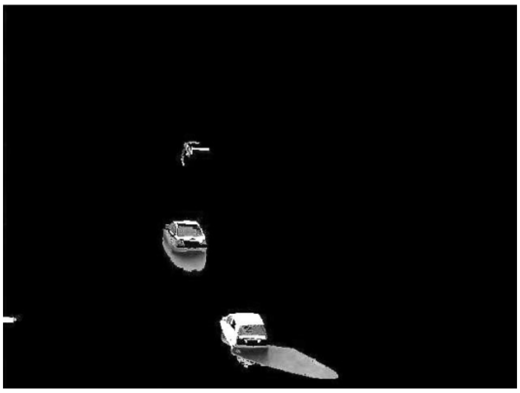


Figure 3. 17: Brightness distortion of a traffic scene

3.3.1.2 Chromacity Distortion

Chromaticity distortion is defined as the orthogonal distance between the observed color and the expected chromaticity line.

$$CD = || \mathit{Input\ frame} - \alpha.\ \mathit{estimated\ background} || \tag{3.21}$$

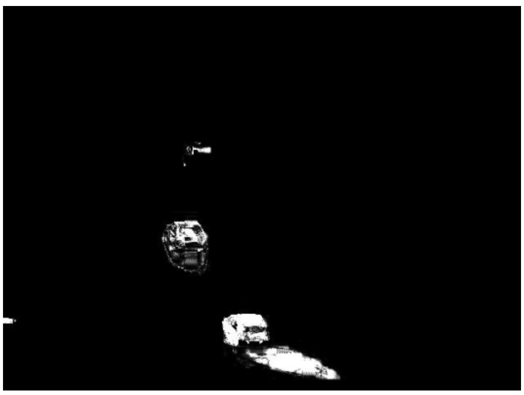


Figure 3. 18: Chromaticity distortion for a sample scene

Given the *RGB* values of a pixel in the input frame as (R_I,G_I,B_I) and background counterpart frame as (R_B ,G_B ,B_B) the brightness distortion, *BD*, can be computed as [24] :

$$BD = \frac{\alpha^2 R_I R_B + \beta^2 G_I G_B + \gamma^2 B_I B_B}{\alpha^2 R_B^2 + \beta^2 G_B^2 + \gamma^2 B_B^2} \tag{3.22}$$

Where α, β and γ are the weights accounting for the influence of *R*, *G* and *B* color components and has been computed making use of *RGB* to *YUV* conversion equations for the luminance component. $Y = \alpha\times R + \beta\times G + \gamma\times B$, with $\alpha = 0.299$, $\beta = 0.587$, $\gamma = 0.114$.

Consequently, a set of thresholds as shown by Table 3.1 are defined to classify the normalized pixels as foreground, shadow or highlights.

Table 3. 1: Shadow and highlight detection thresholds

```
If CD < 10 then
If .5<BD<1 then
        SHADOW
Else if 1<BD<1.25 then
          HIGHLIGT
Else
         FOREGROUND
```

Output obtained by applying the brightness distortion equation is the detected foreground and shadowed background regions. Parts of *FG* object pixels may be misdiagnosed and removed or they may contain some shadow pixels detected as foreground pixels.

3.3.2 Texture Based Analysis

Similar to the color based shadow removal; a texture distortion measure can be defined to detect possible shadow pixels. A simple way of computing the texture is to use the first order spatial derivatives; though other more sophisticated measures can also be employed. We apply Sobel filters in horizontal and vertical directions to both the background and incoming frame and then compute the Euclidean distance between them. If this distance is lower than a certain threshold, *i.e.* very similar texture, then it is highly likely that the pixels are part of a shadowed region.

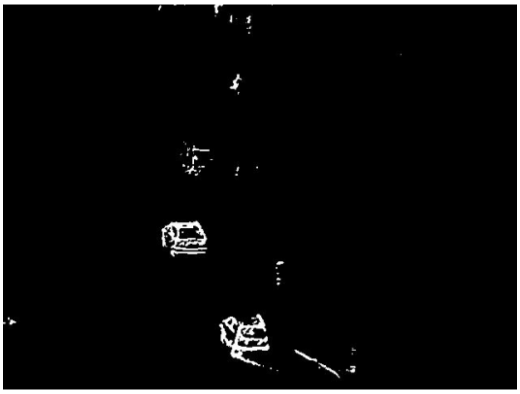


Figure 3. 19: Mask of texture based analysis

3.3.3 Morphological Reconstruction

Mathematical morphological theories can be employed in order to reconstruct the original image without cast shadows or highlights.

By applying *BD* and *CD* Thresholds to the pixels of the input frame, a coarse approximation of foreground object is obtained. While doing morphological reconstruction the estimation of foreground object is called "*Marker*". The so called "*Marker*" image is a binary image where a pixel is set to "1" when it corresponds to a foreground (none cast shadow/highlight pixels). In addition to the marker a "*Mask*" is also generated that contains "1" for pixels which have different texture from the background. These pixels are an approximation of the foreground pixel edge information. A logical *AND* is applied to the Marker and Mask images and the result is edge information of object whose shadows are mostly removed. A morphological dilation is employed to connect vehicle edges together to improve final edge detection of the foreground object.

$$object\_edge = and(mask\ , marker) \oplus SE \qquad (3.22)$$

Here*, SE* is the structuring element for dilation purpose and it depends on the size of foreground to be detected. In this research SE is a box of size (3,3).

Figure 3. 20: Morphological AND result

After small objects are removed by means of a BWAREAOPEN(.) command in MATLAB, a convex hull approach is used to determine the detected foreground object precisely. Result of applying the above described method to an outdoor scene is depicted in Figure 3.21.

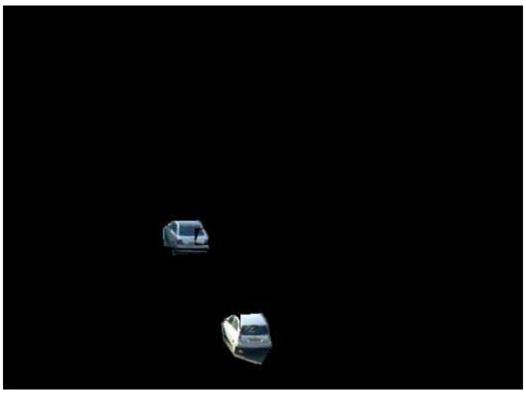


Figure 3. 21: Detected foreground with shadows removed

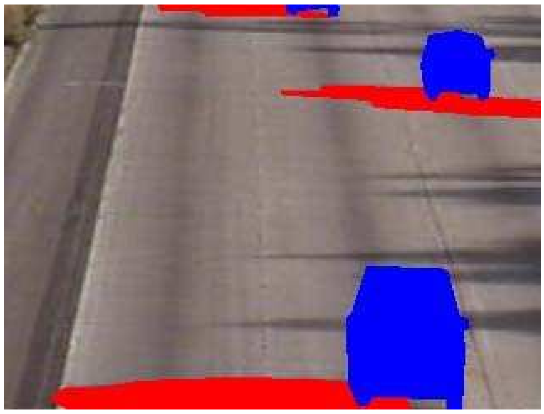
3.4 Evaluation

To have a precise comparison between shadow removal algorithms ground truths video sequences are necessary to provide some quantitative measurements. Although vision based judgment gives a quick idea about the performance of the algorithms it is not an approach that is as accurate as the metric based ones.

3.4.1 Ground Truth Frames

Ground Truth video sequences are generally used for quantitative comparison of different algorithms that try to perform a special task. Video sequences for outdoor traffic are available from Video Surveillance Online Repository (*VISOR*) [56]. In these video sequences some pre-selected frames have the shadow and object pixels pre-identified and represented with different colors in a mask. This enables the researchers to test how well their methods are performing in comparison to the ideal case: in this case the ground truth.

In this study to compare the three different shadow removal algorithms two sets of highway video sequences were used along with their corresponding ground truths. The selected test sequences were Highway I and Highway-II from *VISOR* [56]. A typical frame and its corresponding ground-truth from Highway-I sequence is shown by Figure 3.22.



(a) (b)

Figure 3. 22: Ground Truth video sequence for shadow removal evaluation, Highway- I video sequence [56]

(a) A typical frame (b) vehicles and corresponding shadow regions

As can be seen from Figure 3.22 (b) the cast shadow regions in the frames have been marked by red color in the entire frame. This color has been specifically chosen so as not to mix with the colors of the recorded scene. This provides the flexibility to simply take the output of the tested shadow detection method and compare the pixels the method selects against the pixels denoted by red. Performance comparisons are done by computing two separate scales that are called precision and recall. The following section gives details about what each scale really means.

3.4.2 Recall & Precision

3.4.2.1 Recall

Recall is a measure of completeness and is defined as number of true positives divided by the total number of elements that actually belong to the foreground- objects (i.e. some of both true positives and false negatives).

$$Recall = \frac{TP}{TP + FN} \tag{3.24}$$

$$Recall = \frac{number\ of\ correctly\ identifiedforeground\ pixels}{number\ of\ foreground\ pixels\ in\ ground\ truth} \quad (3.25)$$

True Positive (*TP*) represents the number of foreground pixels correctly detected by the algorithm. False Positive (*FP*) represent for the number of pixels which are incorrectly classified as foreground objects. False Negative (*FN*) stands for the number of pixels corresponding to shadow pixels which are misclassified as part of foreground object.

3.4.2.2 Precision

Precision is a measure of exactness or fidelity. It is evaluated through dividing the number of items (foreground objects) correctly detected by the total number of pixels classified as foreground by algorithm.

$$Precision = \frac{TP}{TP + FP} \quad (3.26)$$

$$Precision = \frac{number\ ofcorectly\ identified\ foreground\ pixels}{number\ of\ foreground\ pixels\ detected\ by\ algorithm} \quad (3.27)$$

3.4.3 Data Analysis

The precision and recall values obtained by the evaluation of the shadow detection routines indicate that shadow removal in *HSV* color space is the most precise method. The precision value of the method means that 82.59 % of the detected foreground objects pixels after shadow removal are actual foreground pixels.

Table 3. 2: Recall and Precision for different shadow removal algorithms

	Precision	Recall
HSV Color Space Based	82.59 %	50.88 %
SCS Based	63.28 %	66.29 %
Color & Texture Based	68.34 %	53.86 %

CHAPTER 4

LICENSE PLATE RECOGNITION

4.1 Introduction

Automatic license plate recognition (*LPR*) plays an important role in detecting vehicles disobeying red lights at junctions. A plate is a label attached to vehicles in order to distinguish them according to certain rules of each government. In general license plates consist of alphabetical characters and numbers with different sizes and colors.

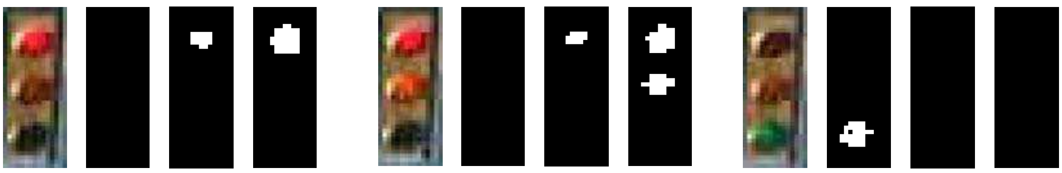
Figure 4. 1:Examples of Mediterranean license plates

License plates in Turkish Republic of Northern Cyprus (*TRNC*) have five characters, except for taxis and rent cars which have six or seven characters. The first two characters of the license plate are alphabets and the last three characters are numbers. In addition, the front plate is colored in white and black and the rear plate is colored in yellow and black. These features can be utilized during the license plate locating procedure. License plates are divided into two categories: single line and double line of characters. Digits cover most width and heights of the plate region. Every digit is made of a single block. At times there might be some hyphenation characters between the characters and numbers but these are generally ignored at segmentation time. Plate size in *TRNC* for a single line plate is 520 mm × 110 mm and for double line plate is 340 mm × 220 mm. Some rear private vehicle license plates are depicted in Figure 4.2 [55].

Figure 4. 2: Samples for single and double line plates in *TRNC*

4.2 Red Light Tracking and Stop Line Detection

In order to check if there is any red light violators one first needs to determine the state of the traffic lights. If the junction is monitored by a fixed surveillance camera then locating the traffic lights in the current input frame turns out to be a simple crop operation at fixed coordinates. Figure 4.3 below shows three instances of the traffic lights localized in this manner.



(a) (b) (c)

Figure 4. 3: Traffic Lights

(a) Red: stop (b) Red + Yellow: get ready to go (c) green: go

To determine the state of the traffic light in the current frame color information from both the YC_bC_r and *HSV* color spaces can be combined into a number of rules as depicted by equation 4.1.

$$L_{Green} \ : \ Y < 110 \,\&\, C_b < 130 \,\&\, C_r < 116$$
$$L_{Red+Yellow} \ : \ C_r > 170 \tag{4.1}$$
$$L_{Red} \ : \ H > 0.69 \,\&\, S > 0.69 \,\&\, V > 0.69$$

After the color rules are applied and noise like parts are removed connected component analysis is applied to find out how many color have been detected in each column. To make the final decision on the state of the light the logical checks shown below have to be applied.

```
if (cnt_rt == 1 || cnt_gt ==1 || cnt_ryt == 1)
   if (cnt_rt ==1  & cnt_ryt ==1 & sum(sum(gt))<10)
      display('Light is red');
   end
      if (cnt_ryt ==1  &  sum(sum(rt)) == 0)
         display('Light is yellow');
   end
   if (sum(sum(gt)) > 10 & cnt_gt==1)
         display('Light is green');
   end
end
 if (cnt_ryt > 1)
   display('Light is yellow');
end
```

Once the state of the traffic light has been established a test is carried out to see if any of the connected components that are in the foreground mask violate an imaginary stop line or region.

4.3 Algorithm

After a red light violator is detected license plate detection procedure starts. As demonstrated in Table 4.1 the algorithm for detecting a license plate is developed in three main parts. The first part is detection and extraction of the plate from the image containing the vehicle by means of color image processing and plate properties. The second part of the algorithm is to segment the characters of the detected license plate region by means of Connected Component Analysis, color and edge information. Last, a correlation algorithm is used to recognize segmented characters. The method utilizes template characters and compares their properties with segmented character properties and decides on the recognition of each character in turn.

Table 4. 1:License Plate Recognition Steps

4.3.1 License Plate Region Locating

4.3.1.1 Radon Transform

As explained in [69], the Hough transform and the related Radon transform are both able to transform two dimensional images with lines into a domain of possible line parameters, where each line in the image will give a peak positioned at the corresponding line parameters. This has lead to many line detection applications within image processing. In the license plate detection procedure the longest detected line of the foreground is used to find the object's skew angle.

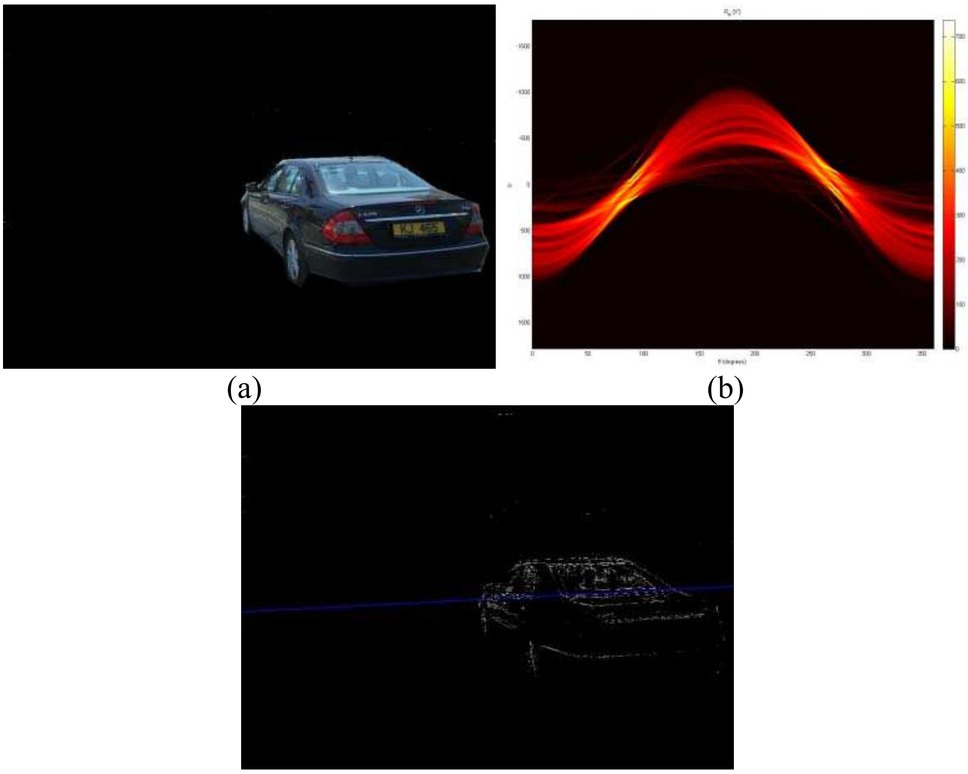
Several definitions of the Radon Transform exist, but a very popular form expresses lines as follows:

$$\rho = x * \cos(\theta) + y * \sin(\theta) \tag{4.1}$$

Where, θ is the angle and ρ the smallest distance to the origin of the coordinate system.

Radon transform for a set of parameters (ρ, θ) is the line integral through the image *g(x,y),* where the line is positioned corresponding to the value of (ρ,θ). The δ is the Dirac delta function which is infinite for argument 0 and zero for all other arguments (it integrates to one).

$$\breve{g}(x,y) = \iint_{-\infty}^{+\infty} g(x,y)\delta(\rho - xcos\theta - ysin\theta)dxdy \tag{4.2}$$



(a) (b)

(c)

Figure 4. 4: Radon Transform

(a) Foreground, (b) Radon transform of the vehicle, (c) Longest detected line

Figure 4.4 depicts an extracted foreground vehicle and its corresponding radon transform. It can be seen from Figure 4.4 (b) that some very bright parts are found in the Radon transform. These parts are the positions of the lines existing in the original image. A simple threshold algorithm could then be used to select longest line of the image and compute its angle with the horizontal axis. Once the skew angle is computed the image can be reverse rotated to make the detected license plate as horizontal as possible.

4.3.1.2 Yellow Region Extraction

To accomplish the first step of the license plate recognition procedure as described in Table 4.1, a combination of color and shape information of license plate is used. After the correction of the skew angle of the detected license plate, an *RGB* to *HSI* transform is employed to help extract all the plate candidates which have yellow color. In the *HSI* color space the yellow color range in Hue channel is as shown in Figure 4.5.

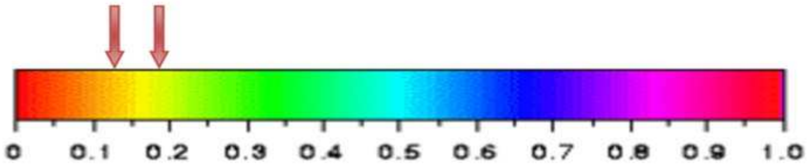
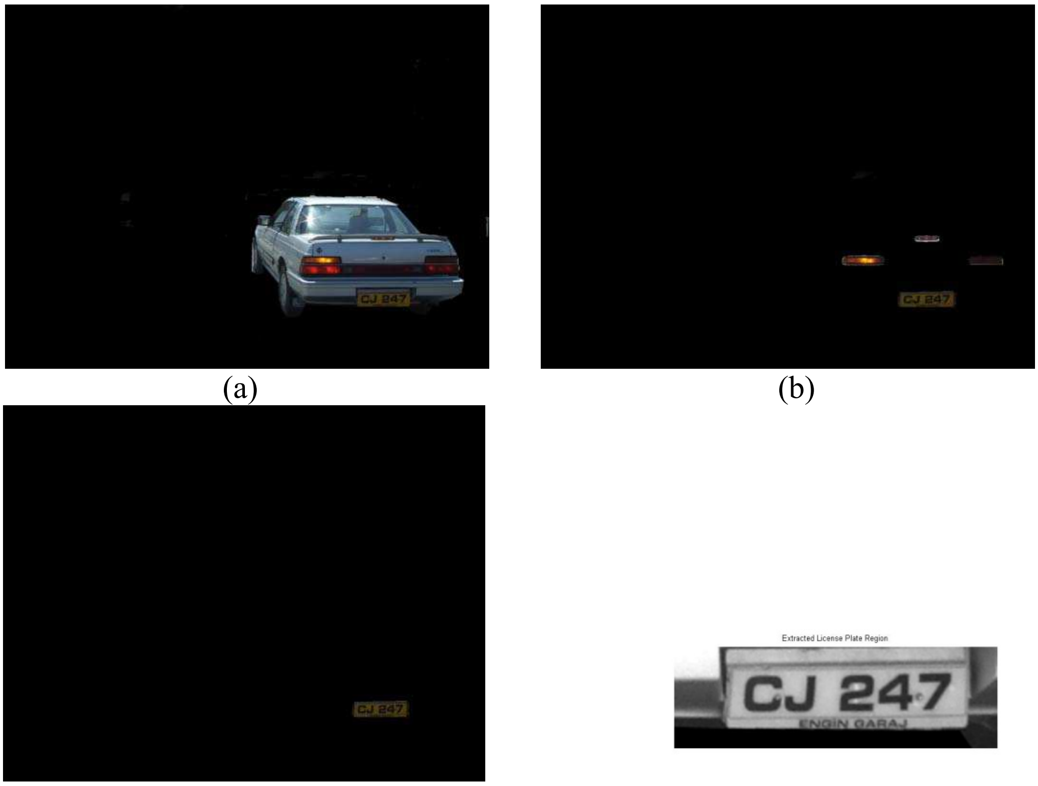


Figure 4. 5: Yellow color pixel range in Hue component of *HSI* color space

Since the aim was to detect red-light violators the frames had been taken from the back side of the cars and as explained earlier rear plates of cars in *TRNC* are yellow. So, plate candidates are regions with yellow color. Subsequently a width to height ratio (*WHR*) is considered to verify regions whose aspect ratios were similar to that of the license plate as shown in Figure 4.6 (b). Finally, an 'area' threshold is used to remove incorrect candidates. Once the correct location of the plate is detected the extracted *RGB* plate image is converted to grayscale by RGB2GRAY(.) command of *MATLAB*. This eliminates the chrominance components and only illumination information will remain.

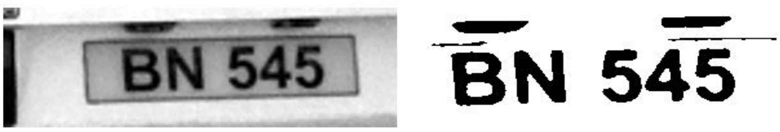


(a) (b) (c) (d)

Figure 4. 6: License plate locating and extraction procedure

(a) Rotated vehicle image (b) width to height ratio filtering (c) locating blob with highest area (d) detected license plate region

To convert the vehicle license plate into a binary image from its gray scale version, an appropriate threshold is employed. After finding the optimal threshold which is based on the intensities of different license plate pixels, binary image with minimum error is obtained.



(a) (b)

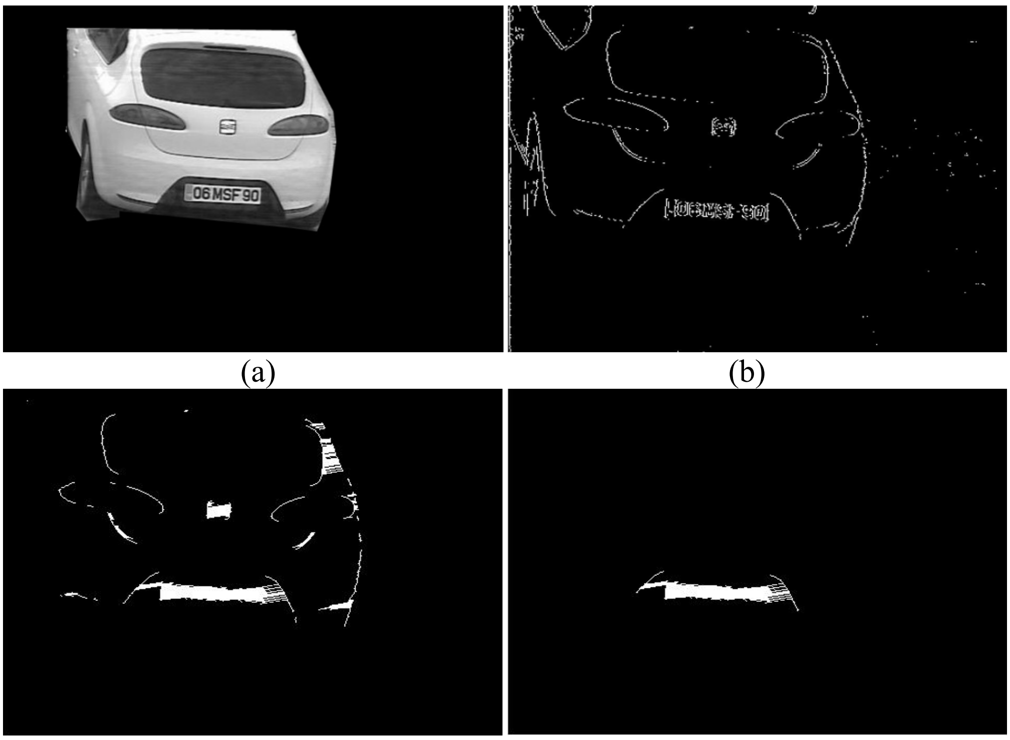
Figure 4. 7: Extracted License Plate Region

(a) Gray level license plate, (b) binary license plate with small errors

Applying some morphological operations final binary version of the license plate can be obtained as depicted in Figure 4.8.

Figure 4. 8: Final version of binary license plate

In case no plate region is detected during color image processing, a vertical edge detection analysis can further be employed on the gray scale version of the foreground image. In this work the Prewitt method of *MATLAB* has been used to find the vertical edges. After discovering the edges, some morphological operations such as dilation and removal of small unwanted objects are applied. These operations aid on finding vertical connected edge areas which are license plate region candidates. Consequently, *WHR* and area thresholds are applied to extract plate region correctly. Important steps are demonstrated in Figure 4.9.



(a) (b) (c) (d)

Figure 4. 9: Vertical edge analysis for license plate detection

(a) An example of gray scale car with white license plate (b) vertical edges of gray scale image (c) Dilated vertical edges obtained from Prewitt Operator (d) detected LP region after applying WHR and Area thresholds

4.3.2 License Plate Character Segmentation

After extracting plate region, next step is to isolate and segment the characters. Character segmentation is a significant step in license plate recognition system. There are many difficulties in this step, such as the influence of image noise, plate frame, rivet, the space mark, plate rotation and illumination variance.

Knowing that all the alphanumeric characters that are used while making plates consists of only one connected component, a connected component analysis (*CCA*) is an efficient method to segment each character or digit. BWLABEL(.) command in *MATLAB* is used to perform *CCA*. Segmented characters for a sample license plate are shown in Figure 4.10.

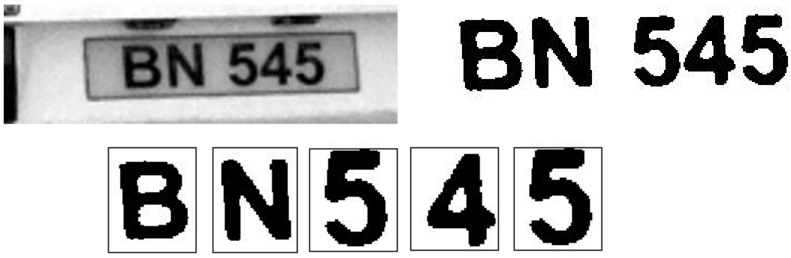


Figure 4. 10: Segmented characters of the license plate

4.3.3 License Plate Character Recognition

License plate characters consist of 10 digits and 24 English characters, totally 34 different characters to be recognized. As the English letters or numbers have only one connected component region, Euler number can be useful to represent the interior structure.

4.3.3.1 Euler Numbers and Characters

Euler number is a structural property of an image. The Euler number or Euler characteristics, is defined as the alternating sum of the number of objects in an image minus the number of holes in it.

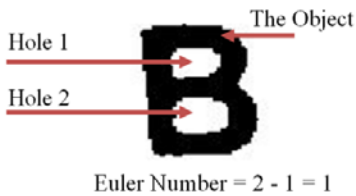
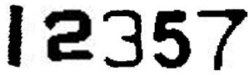


Figure 4. 11: Euler number example

By calculating Euler number of each segment of the license plate image, it is possible to distinguish between six different sets of letters and numbers:

- Numbers without holes (Euler = -1)



- Numbers with one hole (Euler = 0)

- Numbers with two holes (Euler = +1)

- Letters without holes (Euler = -1)

- Letters with one hole (Euler = 0)

- Letters with two holes (Euler = 1)

Placing Euler number filtering before the character recognition procedure improves the accuracy of the character recognition and also at the same time speeds up the processing (fewer comparisons have to be done).

4.3.3.2 Digit Recognition

Before computing any similarity metrics normalization algorithms are applied to both the template and cropped plate characters to make them similar to each other in size. Correlation procedure is done by separating all black and white pixels of both template and cropped plate characters images and counting them separately. Percentage of black and white pixels in both cropped and template characters are calculated. The template character which has minimum difference in the percentage black and white pixels with cropped character is selected to be as the recognized character. Template characters which are used in this thesis are depicted in Figure 4.12.

Numbers:

Letters:

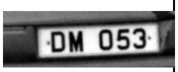
Figure 4. 12: License plate templates of characters and numbers

4.4 Experimental Examples

Some typical test results of practical implementation of the proposed method on LPR in which license plates were successfully recognized are shown in Table 4.2. The method may fail if the video frame is too blurred or if the license plate is highly dirty. Fortunately, pictures were captured by a high resolution camera and they are clear enough to be detected and recognized by the system with a very high resolution rate.

Videos are taken from posterior view of the cars where the license plates are yellow.

Table 4. 2: Practical examples on LPR

Input Frame	Extracted foreground	LP Region	Segmented characters	Recognize d Characters
				GE01 9
				CJ247
				HA335
				DM05 3

4.5 Comparisons with Previous Departmental Works and Thesis Related Publications

Similar works on background estimation and foreground segmentation and violation detection have been done in the electrical and electronics engineering department of Eastern Mediterranean University by Hüseyin Kusetoğulları [72] under supervision of Assoc. Prof. Dr. Hasan Demirel and Sevgin Multlu [71] under supervision of Prof. Dr. Suha Bayındır. Hüseyin has done his research on real time detection and tracking of vehicles for speed measurement and license plate detection. As explained in the thesis photos of the over-speeding vehicles containing their license plate has been captured to be processed later. Their system works in the temporal domain and uses the frames of the acquired video sequences in order to detect the presence or absence of a vehicle moving in the road scene. Inter-frame differencing and background removal methods are employed for the detection of the vehicle in the frames of the video captured by the camera. Background estimation is done by simple method of averaging 10 frames of empty scene. A more complete study on background estimation and subtraction methods has been done by Nima Seifnaraghi in [70] under supervision of Assoc. Prof. Dr. Erhan A. İnce. Nima has applied 6 background estimation/subtraction methods for indoor and outdoor situations. Methods were compared and the fidelity and completeness of the methods versus each other and speed of them in both outdoor and indoor conditions were investigated.

Sevgin has added a laser detector to the vehicle speed and license plate monitoring system to eliminate the disadvantage of the magnetic and fiber optic detectors.

As a result of the research carried out under this thesis two conference publications were made; one in SIU 2009 and the other in ISCIS 2009. A copy of these papers can be found in appendices A and B.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

The main aim of this research was to develop the essential blocks of a system that could detect and identify red light violators in the city by the analysis of surveillance video taken from a fixed video camera. In order to speed up license plate processing and increase the accuracy of license plate detection it was decided that first the foreground containing the red light violator(s) would be separated from the background in the scene. To separate the foreground from the background in the scene a background subtraction algorithm has to be implemented. In this study a recently proposed state of the art *BG* modeling technique known as Group Based Histogram (*GBH*) algorithm has been adopted. The *GBH* is effective and efficient for building a single Gaussian model of background pixels from traffic image sequences. This method is reliable against noise and slow moving objects.

Furthermore to eliminate any cast shadows that may be included in the segmented out foreground three different shadow removal algorithms are introduced and compared against each other with standard and custom taken video sequences. First method for shadow detection was based on the Shadow Confidence Score (*SCS*) computation discussed in chapter 3. Second shadow removal algorithm used the *HSV* color space to eliminate shadow pixels from the extracted *FG*. Final method on shadow removal used a hybrid approach exploiting the color and texture properties of *BG* and input frame to distinguish shaded *BG* from ordinary *BG* or moving *FG* objects. The tests carried out using ground truth frames for the Highway-I test video sequence indicated that the *HSV* based shadow removal would give the best results. In fact the precision value computed for this method was 0.8259 and the next closer method had a precision value of 0.6834.

Before searching for a red light violator in the FG masked input frame, the state of the traffic light should be determined. In this study the traffic lights analysis was carried out using the YC_bC_r and *HSV* color components together with tripwire processing. In Tripwire systems the camera is used to simulate usage of a conventional detector by using small localized regions of the image as detector sites. The simulation results indicate that 99.5% of the time the traffic lights were detected correctly.

Finally, a correlation based *LPR* algorithm was used to recognize the violator's plate number. First the Radon transform was applied to estimate the skew angle of the detected foreground objects and rotation angles were corrected. Then a color and edge based localization for the license plate was carried out. After localization of the plate the individual characters were segmented out using connected component analysis and the characters to test were separated based on their Euler numbers. This was done because Euler number filtering before the character recognition procedure is known to improve the accuracy of the character recognition and also at the same time speeds up the processing. Experimental results indicate that even though the recognition part can further be improved most of the time it is sufficient for correctly recognizing the two letters and three digits on the plates.

5.2 Future Work

Currently only one set of template characters are used while performing correlation based character recognition. However template characters based on different fonts and slant angles could be added to the template set to help increase the recognition accuracy further. Many instances of violations can be recorded (very difficult without breaking the law) and percentage accuracy figures could be provided for correctly identifying the violators.

Also since license plates of taxis and rent cars contain six alphanumeric characters and non-yellow color the routines could be developed to consider these specific instances. The fact that the first character of taxi's license plate is "*T*" and first character of a rent car is "*Z*" can be used in distinguishing these types of plates from the others.

APPENDICES

Appendix A: Novel Traffic Lights Signaling Technique Based on Lane Occupancy Rates

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Abstract—**In a conventional traffic lights controller, the lights either change at constant cycle times or at times proportional to the length of each leg of the intersection. Such approaches clearly are not perfect for optimizing traffic flow. Waiting times proportional to lane length may work well for a single-lane road but when roads with multiple lanes are considered the solution would not be optimal. The authors believe that an adaptive signaling based on fullness of each leg of the intersection would be a better approach. This paper presents the segmentation of foreground objects from frames of the surveillance video using an adaptive K-Gaussian mixture model and describes an approach for determining the lane occupancy rates for the north leg of the intersections. To give an accurate fullness measure the cast shadows that might be present in the segmented foregrounds are removed using a combined probability map called the shadow confidence score. Simulation results are provided for two standard and one custom recorded sequence.**

Keywords-component; gaussian mixture model,cast shadow removal, convex hull fitting, convex hull mask, lane occupancy rates

I. Introduction

In visual surveillance applications, a common approach for differentiating moving objects from the static part of the video frames is detection by background subtraction. Background subtraction involves calculating a reference image, subtracting this reference from each new frame and then thresholding the result. The key issue in background subtraction is how to model the background and update the model in order to adapt to the changes of the background. The changes include variations in the intensity, inclination of the incident light(s) and physical changes such as the small motions of background objects (swaying tree branches, moving clouds, rain or snow). Over the years various statistical models have been proposed. For example in [1], Ridder modelled each pixel with a Kalman filter and presented a more illumination robust system. In [2] and [3] it was assumed that the series of intensity values of a pixel can be modelled by a single unimodal distribution. However in time it was shown that a single-mode model will not handle multiple backgrounds well. For modelling of complex and non-stationary backgrounds [4, 5 and 6] suggest the use of generalized mixture of Gaussians (MoG) model. During modelling the pixel distributions can be initialized randomly, using the K-means approximation and the expectation-maximisation (EM) algorithm [7]. Random initialization is known to result in slow learning and at times would even result in instability. Initialization with the K-means or the EM algorithm would give significantly better results. The EM algorithm is computationally intensive and takes the initialization process off-line. In this study since we would be dealing with real time video from a busy plaza (many moving humans and vehicles) the on-line K-means initialization was adopted. In [8] Kim reported that for backgrounds with fast variations, the MoG with multiple distributions will not be accurate enough and suggested that for such situations non-parametric techniques [9] are used. However in this study since the observed location is an intersection no such fast variations will be encountered (generally people slow down as they approach the traffic lights, and when the light turns green they start from stationary position and gradually speed up).

Various results presented in the literature point out that neither motion segmentation nor change-detection algorithms can distinguish between moving objects and moving shadows. In order to guarantee accurate segmentation shadow suppression must be administered. Shadows occur when objects partially or fully block direct light coming from a source. They are composed of self-shadow and cast shadow. The former is due to the fact that a part of the object is not illuminated directly by the light source and the latter is the region projected by the object in the direction of light. Many shadow removal algorithms exploiting the variations of the brightness and chrominance distortion metrics (*BD*, *SBD*, *HBD*, α_i, CD_i), the *YUV* or *HSV* colour spaces, and texture difference between foreground and background exist. In this paper for the detection of cast shadows the total shadow confidence score proposed by Fung in [11] will be used.

II. Adaptive K-Gaussian Mixture Model

As described in [4] and [5] each 3-tuple pixel vector in the current scene is modeled by a separate mixture of K Gaussians:

$$P(X_{i,t}) = \sum_{i=1}^{K} w_{i,t} \cdot \eta(X_{i,t}, \mu_{i,t}, \Sigma_{i,t}) \quad (1)$$

In the adaptive *K-MoG* model $X_{i,t}$ is the current pixel value vector which consists of *R*ed, *G*reen and *B*lue components, $w_{i,t}$ is an estimate of the weight of the i^{th} Gaussian in the mixture at time *t*, $\mu_{i,t}$ and $\Sigma_{i,t}$ are the mean value and the covariance matrix of the i^{th} Gaussian in the mixture. $P(X_{i,t})$ denotes the probability of observing the current pixel value vector given the mixture of *K* Gaussian distributions and $\eta(X_{i,t}, \mu_{i,t}, \Sigma_{i,t})$ is a Gaussian probability density function.

$$X_{i,t} = (x_{i,t}^R, x_{i,t}^G, x_{i,t}^B)$$
$$\mu_{i,t} = (\mu_{i,t}^R, \mu_{i,t}^G, \mu_{i,t}^B)$$
$$\Sigma_{i,t} = \begin{pmatrix} \sigma_R^2 & 0 & 0 \\ 0 & \sigma_G^2 & 0 \\ 0 & 0 & \sigma_B^2 \end{pmatrix} \quad (2)$$
$$\eta(X_{i,t}, \mu_{i,t}, \Sigma_{i,t}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(X_{i,t}-\mu_{i,t})^T \Sigma^{-1}(X_{i,t}-\mu_{i,t})}$$

Background/foreground separation consists of two independent steps: 1) estimating the parameters of *K* distributions; and 2) evaluating the likelihood of each distribution to represent the background.

A. Parameter Updating

Since at the start of modelling all the Gaussians have an equal probability for representing the background the weights $w_{i,t}$, i∈ (1...K), are all set to the value 1/*K* and the variances are set randomly to high values. Then every new pixel value vector $X_{i,t}$ is checked against the existing *K* Gaussian distributions until a match is found (a match is defined as a pixel value vector whose Euclidean distance is within 1.5 standard deviations of a distribution). The parameters of the matched component are then updated using the recursive equations below:

$$\mu_{i,t} = (1-\rho) \cdot \mu_{i,t-1} + \rho X_{i,t}$$
$$\Sigma_{i,t} = (1-\rho) \cdot \Sigma_{i,t-1} + \rho \cdot diag\{(X_{i,t} - \mu_{i,t})^T (X_{i,t} - \mu_{i,t})\} \quad (3)$$
$$\rho = \alpha \cdot \eta(X_{i,t} | \mu_{i,t-1}, \Sigma_{i,t-1})$$

In equation (3) α represents the user-defined learning rate and has a value in the range $0 \leq \alpha \leq 1$. ρ on the other hand is a learning rate for the parameters.

For the case when there are no matches the Gaussian distribution with the least weight is replaced by a new component with a mean equal to the current pixel vector. The variance for this new distribution is set high and the weight is set to a low prior value. Finally, the weight of all the *K* Gaussians (G_i, $i \in 1...K$) at time *t* are updated and normalized using equation (4)

$$w_{i,t} = (1-\alpha) \cdot w_{i,t-1} + \alpha \cdot M_{i,t}$$
$$w_{i,t} = \frac{w_{i,t}}{\sum_{m=1}^{K} w_{m,t}} \quad (4)$$

When there is a match $M_{i,t}$ is assumes as 1 and 0 otherwise.

B. Background Estimation

Once the parameters for all the Gaussian distributions are updated the ones that are most likely produced by background processes are determined. First, the *K* Gaussians are sorted in descending order by the value of $w_{i,t} / \Sigma_{i,t}$ and then the first *B* distributions are chosen to be in the background model using the value of *B* as given by (5)

$$B = \arg\min_b \left\{ \sum_{k=1}^{b} w_k > T \right\} \quad (5)$$

Here, *T* assumes a value between 0.5 and 1. Generally the segmented foreground would contain some noise. It is possible to get rid of this noise by making use of morphological operations and connected component analysis [13].

III. CAST SHADOW DETECTION AND REMOVAL

Ideally, background subtraction should detect real moving objects with high accuracy. However in practice the detection of cast shadows as foreground objects is very common. The cast shadows that are projected on the road surface can change in size based on the elevation of the illuminating light source. When cast shadows stretch, two or more independent objects can appear to be connected together. Unavoidably, the accuracy of segmentation and estimation of how full the intersection legs are would be affected negatively. In order to alleviate these problems the paper adopts a combined probability map also known as the shadow confidence score (*SCS*) [10,11]. The characteristics of the cast shadow in the luminance, chrominance and gradient density domain dictates that:

- Luminance values of the cast shadow pixels are lower than those of the corresponding pixels in the background image.
- The chrominance values of the cast shadow pixels are identical or only slightly different from those of the corresponding pixels in the background
- The difference in gradient density values of the cast shadow pixels and the corresponding background pixels is relatively low. The difference in gradient density values between the vehicle pixels and the corresponding background pixels is relatively high

Based on these observations, the luminance, the chrominance and the gradient density scores for each blob in the foreground mask can be computed using the equation defined in [11] and a total shadow confidence score (*SCS*) can be obtained by combining these three scores.

Figure 2 depicts the effect of *SCS* based shadow removal for frame #1590 of the custom video. From the *SCS* depicted in part (d) it can be seen that the pixels representing the foreground objects in comparison to the cast shadows have lower intensities. To distinguish between the two a threshold may be applied and as depicted in Figure 2(e) sometimes parts of the objects can be misclassified as shadows (incorrect decisions led to undesired erosion on the foreground mask). To fix this problem a convex hull can be fitted to the remaining shadow free foreground mask and then inside the hull is filled to create a new more complete foreground mask. Finally the convex hull based new mask can be used to segment the foreground objects from the input frame.

A. Convex Hull Fitting

Generating a polygon that completely and closely surrounds a given set of points in 2D is called convex hull fitting. In the literature there are many algorithms for convex hull generation. Some well-known ones include incremental, gift wrapping, divide and conquer and quick hull algorithms. In this paper we describe the incremental algorithm. The processing starts with a single point and then using two more points a triangle is created. Next a new point is selected. If the new point is inside the hull there is nothing to do. Otherwise one must delete all the edges that the new point can see and add two new edges to connect the new point to the remainder of the old hull. This process is then repeated for all the remaining new points.

B. Convex Hull Mask

As mentioned earlier while trying to separate objects from their cast shadows some pixels belonging to the actual vehicles can be misclassified as shadow and this would cause partial erosion or holes to appear on the foreground objects mask. Creating a new mask by using the set of points included in the convex-hull would bring a solution for this problem. As can be seen from Figure 1 every two points in the convex hull (red stars) can define a line and when all the lines are considered we have a closed polygon. The new mask will be composed of all the points that fall inside this polygon.

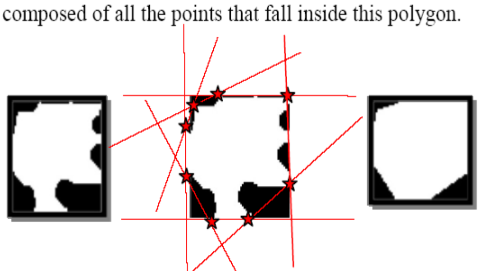

Figure 1 – Creation of convex-hull mask

Figure 2(g) and 2(h) shows the newly created convex hull mask and the segmented *RGB* foreground objects based on this new mask.

IV. TEST SEQUENCES

The background estimation and shadow removal algorithms discussed above were tested using two standard sequences (*Highway-I* and *Highway-II*) and one custom recorded video sequence. *Highway-I* sequence has a frame rate of 25 samples per second and a resolution of 352×288. *Highway-II* which is obtained from VISOR image lab [14] has a frame rate of 15 samples per second and a resolution of 320×240. The custom sequence was recorded by a Fuji Film Fine Pix S6000fd camera. The recording speed was set at 30 frames per second and the resolution was 680×480.

V. SIMULATION RESULTS

In this study the adaptive *K*-Gaussian mixture modeling was not applied to every single pixel in the input frame. Instead we looked at the difference between the current frame and a previous reference frame and if the difference was less than a previously defined threshold (35 in this study) the tested pixel was assumed to be part of the background. The adaptive model was only applied for the locations where the difference was more than the selected threshold. It was found that this would greatly speed up the processing.

For all video sequences K=7, α=0.05, β=1.5, and T =0.85 were used in the adaptive *K-MoG* model. The thresholds used by the *SCS* calculator have also been summarized in Table 1. While for the higher resolution custom video a separate set of threshold values were required for the standard sequences same set of values was sufficient.

TABLE I. SHADOW DETECTION ALGORITHM PARAMETERS

Yeni-Izmir Junction	TL=180	TC1=9.5	TC2=19	TG1 = 0.3	TG2 = 0.6
Highway-I	TL=200	TC1=7.5	TC2=15	TG1 = 0.5	TG2 = 1.0
Highway-II	TL=200	TC1=7.5	TC2=15	TG1 = 0.5	TG2 = 1.0

As pointed out in section 3 the first set of results were obtained using the *Yeni-İzmir* custom sequence. These results are presented in Figure 2. The second set of simulations were carried out using the standard *Highway-I* sequence. Figure 3 shows the results of background estimation, subtraction, and shadow removal steps as applied to frame #1230. Figure 3(c) depicts the extracted foreground with some shadow, (e) is the computed shadow confidence score, (f) shows the remaining foreground after shadow is removed, (g) is the convex hull based new mask and (h) is the foreground objects with minimal shadow.

In [12], an edge-based moving shadow removal algorithm composed of many steps (high computational complexity) had been proposed and the authors had claimed that their approach would give better results than the ones obtained in the *HSV* domain or than by using the *SCS* based map. This statement would only be true if the foreground segmentation is done right after SCS is thresholded. If after the application of the threshold a convex hull is fitted to the partly deformed foreground and a new mask based on the points defining the convex hull is created then the segmented foregrounds using this new mask would be as good as the ones obtained in [12]. Simulation results shown in Figures 2, 3 and 4 all indicate that segmentation with a convex hull based mask works quite well.

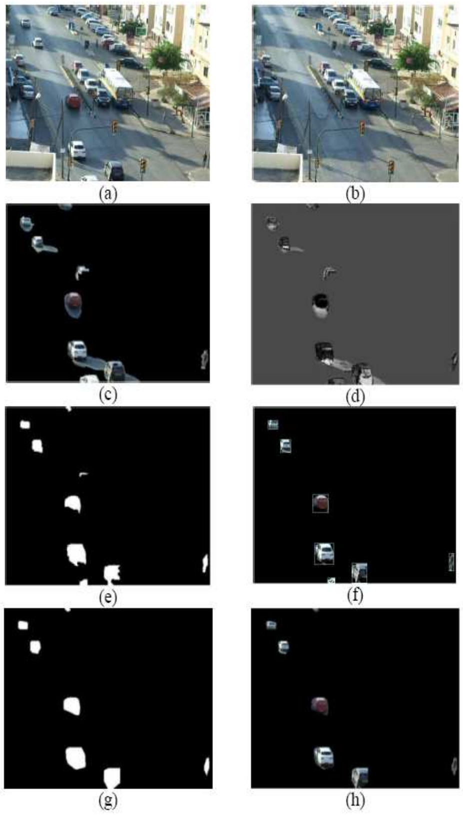

Figure 2 – Foreground segmentation and shadow removal for Yeni-İzmir Junction

(a) input frame (b) estimated background (c) RGB foreground with shadows (d) total SCS (e) shadow free foreground mask (f)bounding box cropped FG (g) new convex hull mask (f) FG objects cropped by new convex hull mask.

A. *Analysis of Lane Fullness*

Assuming that in real life each leg of an intersection is being monitored simultaneously by fixed surveillance cameras this section suggests a way for computing the fullness of a single leg of an intersection. In systems using fuzzy logic each leg houses two sensors behind traffic lights separated by a distance D. The sensor at distance D from the light counts the number cars coming to the intersection and the second counts the cars passing the traffic light. The amount of cars between the sensors is determined by the difference of the readings. However, this approach can not differentiate between a truck, a bus or a car. Hence determining what percent of the road is full based on size becomes fairly difficult.

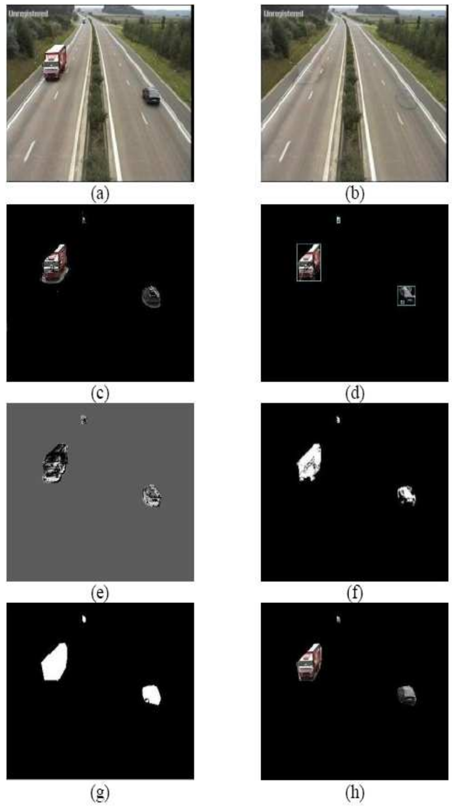

Figure 3– Segmentation and shadow removal for Highway-I

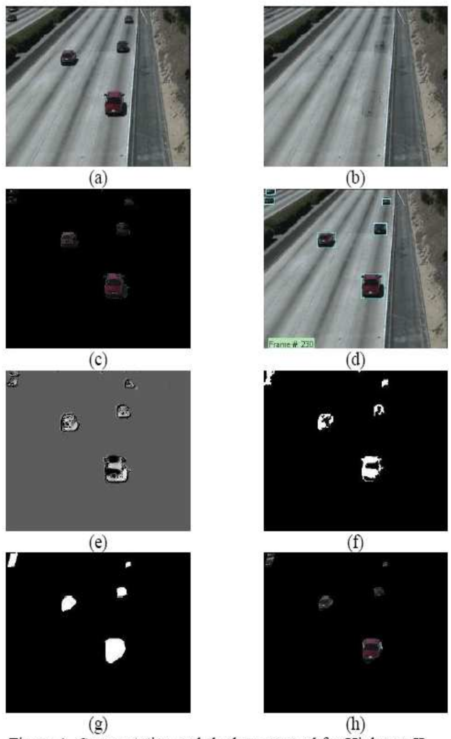

Figure 4– Segmentation and shadow removal for Highway-II

A better approach that would not require any information on the type of cars present behind the traffic lights would be the use of the foreground mask(with shadows removed) together with two lane masks for determining how much each lane and the detected foreground overlap outside a designated region *A* (ref to Figure 5(e)) . Afterwards we test to see if any of the foreground objects fall in this designated region. If region *A* contains no moving objects it is assumed 100% full. Otherwise the overlap between the extracted FG over region A and the ground truth mask of region A is computed. The application of the fullness analysis to the north leg of the intersection for frame #1890 is depicted in Figure 5.

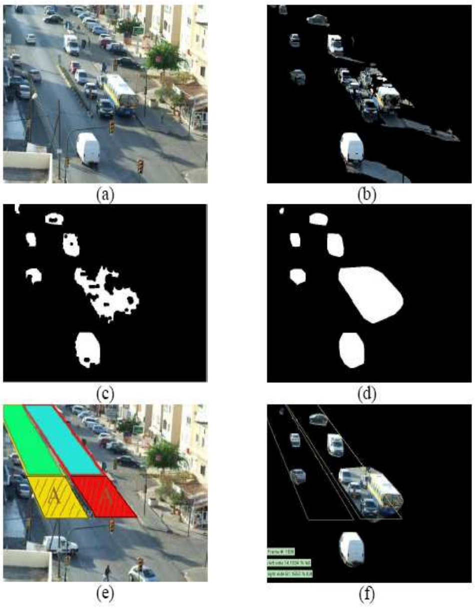

Figure 5– Lane masks and fullness analysis

VI. CONCLUSIONS

The paper proposed an adaptive signaling technique based on fullness analysis of the different legs of an intersection and gave an example for the north leg of *Yeni-İzmir* Junction of Famagusta. It also introduced the concept of applying a convex hull on the output obtained from shadow removal routines operating in *HSV* domain or using a combined probability map known as *SCS*. Various examples were provided using standard and custom sequences to show the advantage of applying the convex hull before segmentation. Future work will include the collection of all the fullness measures for the four different legs and signaling of the control device to switch the lights based on the analysis of the collected data.

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Appendix B: Traffic Analysis of Avenues and Intersections Based on Video Surveillance from Fixed Video Cameras

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Özetçe

Bu makalede sabit bir görsel gözetim sistemi tarafından gözetlenmekte olan herhangi bir cadde, kavşak veya seçilmiş bölgedeki hareketli nesnelerin uyarlanır Gauss Fonksiyonları Karışımı (GFK) yöntemi kullanılarak arkaplandan ayrıştırılması ve kavşak kollarındaki doluluk oranı analizleri sunulmaktadır. Kavşak kollarında veya caddelerdeki doluluk oranlarını doğru kestirebilmek için öncelikle ön-plandaki gölgelerin mümkün olduğunca elimine edilmesi gerekmektedir. Bu çalışmada gölge kestirimi ve silinmesi renk özü, doygunluğu ve yeğinliği uzayı (HSV) kullanılarak gerçekleştirilmiştir. Benzetimler PETS 2001 Camera 1 dizini ve KKTC-Mağusa şehrinde kaydedilen bir dizinin kullanımıyla gerçekleştirilmiştir. Kavşak bacaklarının sağ ve sol şeritlenin yüzde cinsinden doluluk hesaplaması için yeni bir yöntem önerilmiş ve bu oranlar seçilmiş çerçeve üzerine işlenmiştir.

Abstract

Based on adaptive Gaussian mixture modelling this article presents the separation of foreground objects from frames of surveillance video taken at avenues and/or intersections. The paper also describes an approach for determining the lane fullness of a dedicated leg of an intersection. In order to give an accurate fullness measure the cast shadows that might be present in the segmented foregrounds must be eliminated. In this study the detection and removal of shadows have been carried out using the HSV color space. The simulations were carried out using the Camera 1 sequence from PETS 2001 database and a custom sequence recorded in TRNC-Famagusta. A new method for computing right and left lane fullness in each leg of the intersection has been proposed and values computed have been recorded on the bottom left corner of the frame under study.

1. Giriş

Görsel gözetlemede hareketli nesneleni arka plandan ayırmanın yaygın yolu arkaplan kestirimi ve mütakiben arkaplan çıkarımıdır. Bazı yöntemler bir pikselin zamana yayılmış seri halindeki yeğinlik değerlerinin tek doruklu bir dağılım fonksiyonu kullanılarak modellenebileceğini varsaymaktadır [1], [2]. Bununla birlikte tek doruklu bir model sallanan ağaç dalları veya savrulan kar zerreciklerinin neden vereceği çoklu arkaplanlarla başa çıkamamaktadır. Genellikle, karmaşık ve durağan olmayan arkaplanların modellenmesi işinde genelleştirilmiş Gauss Fonksiyonları Karışımı (*GFK*) kullanılmaktadır [3],[4],[5]. Modelleme esnasında ilklendirme ve parametre güncelleme beklenti enbüyütme yöntemi (*EM*) [6] veya *K*-ortalama yöntemi ile gerçekleştirilebilir. *EM* ilklendirmesi kullanan yöntemlerin daha iyi sonuç verdiği bilinse de bu yaklaşımın karmaşıklığı ve başlangıçta çevrimdışı olmasından dolayı bu çalışmada K-ortalama ilklendirmesi kullanılmıştır. Modelleme esnasında zeminde hızlı değişimler oluyorsa *GFK*'nın bile 3-5 adet Gauss fonksiyonu ile yeterince doğru bir sonuç vermeyeceği [7] de belirtilmektedir. Bu problemleri çözecek parametrik olmayan ve çekirdek yoğunluk kestirimi kullanan bir yöntem [8] de sunulmuştur. Bu çalışmada görsel sistemin kaydettigi çerçevelerdeki zeminde hızlı ve ani değişiklikler olmadığından parametrik olmayan yöntemlerin kullanımına ihtiyaç olmamıştır.

Bildiri düzeni aşağıdaki gibidir. İkinci kısım Gauss Fonksiyonları Karışım Modeli ve bu modeldeki parametre güncelleme detaylarını anlatmaktadır. Daha sonra 3. kısımda önplan/arkaplan ayrıştırma işlemi sonunda ön planın bir parçası olarak bulunan gölgelerin *HSV* uzayında kestirimi ve silinmesi anlatılmıştır. Dördüncü bölüm kullanılan video test dizinlerini ve bu dizinlerin özelliklerini belirtmiş, beşinci kısımda ise benzetim sonuçları sunulmuştur.

2. Gauss Fonksiyonları Karışım Modeli

Uyarlanır Gauss fonksiyonları karışım modeli [3] ve [4] de belirtildiği üzere her çokuzlu piksel vektörünü *K*-adet Gauss dağılımı karışımından oluşacak şekilde modellemektedir. Çokuzlu piksel vektörü X_t kırmızı, yeşil ve mavi bileşenlerden oluşan bir yeğinlik değerler vektörünü, $w_{i,t}$ belirli bir zamanda karışımdaki her Gauss dağılım fonksiyonu için kestirilmiş bir katsayıyı, $\mu_{i,t}$ ve $\Sigma_{i,t}$

ise karışımdaki her dağılım fonksiyonunun avaraj değer ve ortak değişinti matrisini temsil etmektedir.

Ön/arka plan ayrıştırma işlemi esasen iki bağımsız problem olarak görülebilir. Bunlardan ilki K elemanlı karışımdaki Gauss fonksiyonları ile ilgili parametrelerin kestirimi, ikincisi ise her dağılımın arkaplanı temsil etme olasılığının değerlendirilmesidir.

2.1 Parametre Güncellenmesi

Başlangıçta tüm Gauss fonksiyonları eşit olasılıklı olduğu için tüm katsayılar, $w_{i,t}$, $1/K$ değerine eşitlenmekte ve değişinti değerleri de rastgele yüksek değerler olarak alınmaktadır. Daha sonra her çokuzlu piksel vektörü eldeki K-adet Gauss fonksiyonu ile karşılıştırılmaktadır. Bir çakışma durumunda (çokuzlu piksel vektörü X_t'nin herhangi bir dağılımdan 2.5 standart sapmadan daha yakın olma durumu) ilgili dağılım ve/veya dağılımların parametreleri güncellenmektedir.

Tarama sonucunda bir çakışma bulunmaması durumunda katsayısı en düşük olan Gauss dağılımı bir yenisi ile değiştirilir. Bu yeni dağılımın katsayısı düşük, değişintisi yüksek ve avaraj değer vektörü ise X_t ye eşit tutulur. Gauss dağılımına ait katsayılar ise her gözlemleme zamanı için değiştirilmektedir. Çakışma var ise $M_{i,t}$ 1 diğer durumlar için 0 olarak kabul edilmektedir.

2.2 Arkaplan Kestirimi

Her çokuzlu piksel vektörü için parametre güncellemesi yapıldıktan sonra dağılımlar w_k / σ_k değerlerine göre sıralanmakta ve üst tarafta kalan B adet dağılımın zemini en iyi temsil eden dağılımlar olduğu kabul edilmektedir.

$$B = \arg\min_{b} \left\{ \sum_{k=1}^{b} w_k > T \right\} \qquad (1)$$

Denklem (1)'deki T eşik değeri 0.5 ile 1 arasında değişebilmektedir.

2.3 Morfolojik İşlemler

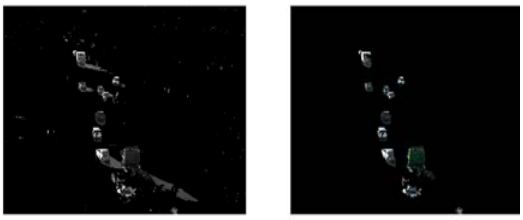
Üzerinde çalışılan çerçeve ön ve arkaplan olarak ayrıştırıldıktan sonra genelde ön-planda hareketli nesneler ile birlikte bazı gürültüler de bulunur. Bu gürültüleri mümkün olduğunca azaltmak için bazı morfolojik operatörlerden yararlanılmaktadır. Şekil 1 de ayrıştırma sonrası elde edilen bir ön plan görüntüsündeki gürültünün nasıl minimize edildiğini gösterilmiştir. Gürültüden kurtulmak için hem kapama ve açma işlemleri hem de bağlantılı bileşen analizi uygulanmış ve piksel sayısı toplamı belli bir eşik değerin altında olan bileşenler gürültü kabul edilip ön plan çerçeveden silinmiştir.

3. Gölgesizleştirme

Günün değişik saatlerinde cadde üzerinde hareketli haldeki nesneler belli açıdan vuran ışığı bloke edeceği için yol üzerinde gölgeler oluşacaktır. Bu gölgeler ışık kaynağının yüksekliğine göre uzayabilmekte veya daralabilmektedir. Gölgelerin uzaması durumunda birbirinden bağımsız iki nesne birleşebilmekte ve bu hem araç sayımını hem de şerit doluluk oranı hesaplarını olumsuz yönde etkilemektedir. Bu yüzden çalışmamızda [10] da belirtilen renk özü, doygunluğu ve yeğinliği uzayını kullanan bir gölge nokta maskesi (*GNM*) ve deneysel olarak elde edilen bazı eşik değerleri gölgesizleştirme amaçlı kullanılmıştır. Bu çalışmada gölge sezim ünitesince 'ht'ya. duyulan α, β, τ_S ve τ_H değerleri sırası ile 0.48, 0.95, 0.4 ve 0.7 olarak alınmışlardır.

$$GNM_k(x,y) = \begin{cases} 1 & \left\{ \alpha \le \dfrac{I^V{}_k(x,y)}{B^V{}_k(x,y)} \le \beta \;\cap \right. \\ & \left(I^S{}_k(x,y) - B^S{}_k(x,y)\right) \le \tau_s \cap \\ & \left. \left|I^H{}_k(x,y) - B^H{}_k(x,y)\right| \le \tau_H \right\} \\ 0 & \textit{aksi takdirde} \end{cases} \qquad (2)$$

Denklem (2) de $I_k(x,y)$ ve $B_k(x,y)$ giriş video dizini ve arkaplan modellerinin k'inci çerçevesindeki (x,y) koordinatlı piksel değerlerini temsil etmektedir.



(a) (b)

Şekil 1: Magosa_Yeni_İzmir_Kavşağı, çerçeve 2100 de gürültüsüzleştirme ve gölgesizleştirme.

4. Video Test Dizinleri

Yapılan çalışmaların performans değerlendirmesi için PETS 2001 [11] veri tabanından Kamera 1 ve Kamera 2 dizinleri ile Magosa Yeni İzmir kavşağında AVI olarak kaydedilmiş "Magosa_Yeni_İzmir_Kuzey" dizini kullanılmıştır. Fuji S6500 marka kamera ile kaydedilen dizin 640×480 çözünürlük ve saniyede 30 çerçeve içermektedir. Kamera kayıt yaparken hareketli *JPEG* sıkıştırması uyguladığından çerçevelerin MATLAB ortamına okunabilmesi için sıkıştırmayı açacak bir kod çözücüsü yüklenmiştir.

5. Benzetim Sonuçları

Bu çalışmada video dizinlerindeki her çerçeveye Gauss fonksiyonları karışım modeli uygulanmadan önce bir eşikleme uygulanmıştır. Güncel çerçevedeki (x,y) koordinatlarındaki piksel geçmişle kıyaslandığında oldukça uzun bir süre değişmemişse bu piksel sabit varsayılmaktadır. Bir başka deyişle güncel çerçeve ile önceki bir referans çerçevesindeki değer farkı deneysel olarak seçilmiş bir eşik değerinin (minDiff) altında ise bu piksel arkaplanın bir parçası olarak kabul edilmekte ve uyarlanır süreç atlanmaktadır. Bu yaklaşım sadece kısıtlı sayıdaki koordinatta uyarlanır süreci kullanacağı için hem çerçeve hem de dizinin çok daha hızlı işlenmesine neden vermektedir.

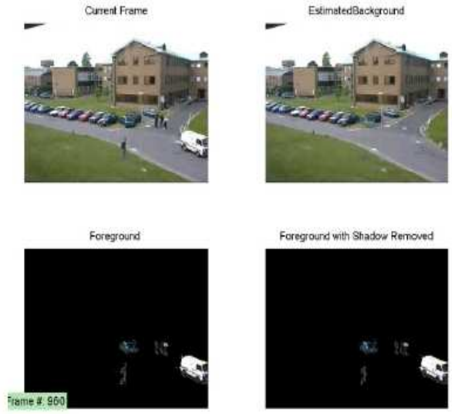
Önceki bölümde belirtilen test video dizinleri için bu çalışmada seçilen eşik değeri 35dir. 5-bileşenli *GFK* modelinde $\alpha = 0.05$, $\beta = 2.6$ ve $T = 0.85$ olarak alınmıştır. İlk deneme PETS-2001 veritabanından Kamera 1 test dizinine uygulanmış ve geliştirilen algoritma arkaplan kestirimi ve ayrıştırmayı başarıyla gerçekleştirmiştir. Şekil 2 bu dizinin 960'ıncı çerçevesi için ayrıştırma, gölgesizleştirme ve hareketli nesnelerin belirlenmesi işlerini göstermektedir. Hareketli nesnelerin birbiriyle fiziki olarak örtüştüğü durumlarda nesneleri ayırma işlemine gidilmemiştir.

İkinci deneme Magosa şehrinde Yeni İzmir kavşağında çekilen bir video dizini kullanılarak gerçekleştirilmiştir. Bu dizin yakın çekimle kavşağın kuzey bacağını görüntülemektedir. Güneşli bir günde ve öğle vaktinden önce çekilen bu dizinde hareketli cisimlerin ışığı bloke etmesiyle yol üzerinde oluşmuş değişik boylarda gölgeler mevcuttur. Bu gölgeler hem nesnelerin ön planda birleşmesine hem de hareketli nesneleri belirlerken yanlış kabullere neden verebilmektedir. Gölge sezim ünitesinde α, β, τ_S ve τ_H değerleri sırası ile 0.48, 0.95, 0.4 ve 0.7 olarak alındığında Şekil 3 deki sonuçlar elde edilmiştir. Çalışma yanlış kabulleri minimize etmek amaçlı olarak hareketli nesne bulunduğu varsayılan bölge ile arkaplan resmindeki ayni bölge arasındaki ilinti değerlerini de hesaplamış ve bu değerin yüksek olduğu durumlar için o bölgeyi seçmekten kaçınmıştır.

5.1 Şerit Doluluk Analizi

Geleneksel trafik ışık sistemleri ya eşit ya da her şeridin uzunluğu ile orantılı bekleme peryotları gerektirmektedir. Bu tür sistemler ve uyguladıkları mantık ile trafik akışının optimize edilmesi yetersiz kalmaktadır. Dörtyolların değişik bacaklarında bekleyen araçlara uyarlanır işaretleşmeye dayalı yol vermenin (ihtiyaca ve doluluk oranına bağlı olarak) tüm yönlerdeki trafik akışını optimize edeceği düşünülmektedir.

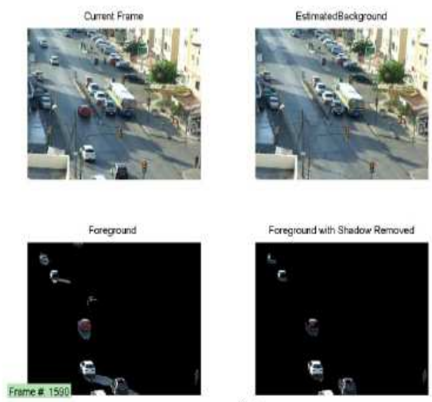
Denetim altındaki kavşağın tüm bacaklarının tek bir güvenlik kamerası ile izlenmesi mümkün olmadığından bu çalışmada sadece kavşağın kuzey bacağı için doluluk oranı



(a)

(b)

Şekil 2: PETS 2001 Kamera 1 dizininin 960'ıncı çerçevesi



(a)

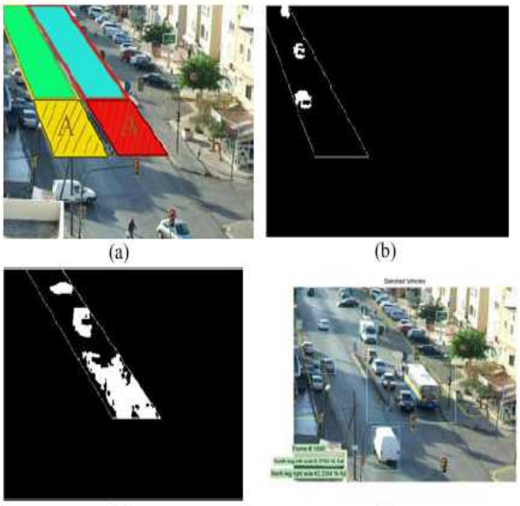
(b)

Şekil 3: Magosa_Yeniİzmir_Kuzey dizinindeki 1590'ıncı çerçevede önplan ayrıştırma , gölgesizleştirme ve hareketli nesne tesbiti.

hesaplamaları verilmiştir. Şekil 4(a)′daki şerit maskeleri ile gölgesizleştirilmiş önplandaki hareketli nesnelerin yüzde kaç örtüştüğünü (şerit doluluk oranı) tespit edilerek her şeridin yüzde doluluk oranı bulunabilmektedir. Trafik ışıklarına yakın ön bölgede sınırları belli bir alan seçilmekte (alan-*A*), ve ön/arka plan ayrıştırmasından sonra hareketli nesnelerden herhangi birinin bu alan içinde olup olmadığı sorgulanmaktadır. Seçilmiş bölge içerisine herhangi bir hareketli nesne düşmezse bu bölge 100 % dolu kabul edilmekte geriye kalan yol yüzeyinin doluluk oranı ise önplan ve yol maskelerinin örtüşme oranına göre hesaplanmaktadır.

İnceleme altındaki bacağın her iki şeridinin de doluluk oranı 1890′ıncı çerçeve için Şekil 4 de verilmiştir. Tüm yönlerdeki trafik akışını bu mantıkla kontrol edebilmek için dörtyolun her bacağında ayni analiz eş zamanlı olarak yapılmalı ve kıyaslamalar sonunda kontrol cihazına bir sinyal gönderilmesi gerekmektedir. Eş zamanlı analiz her bacağın ayrı bir *IP*-kamerasıyla izlenmesini şart koşacaktır.

Şerit doluluk analizi yaparken trafik ışıklarına yakın olan, ön bölgede belirlenen alan-*A*′nın derinliği, bu çalışmada sabit tutulmuştur. Fakat video dizininin her 5 dakikalık aralığı taranıp trafik akış hızı belirlenirse bu derinliğin uyarlamalı şekilde değiştirilmesi mümkün olacaktır. Bu da günün farklı zamanlarında yapılan doluluk tahminlerinin daha gerçekçi olmasını sağlayacaktır.



(a) (b) (c) (d)

Şekil 4: Yeni_İzmir kavşağının kuzey bacağındaki sağ ve sol şeritlerdeki doluluk oranı analizi.

6. Vargı ve İleriki Çalışmalar

Bu çalışmada ana caddeler veya herhangi bir kavşak üzerindeki nesnelerden hareketli olanların arkaplan kestirimi ve ön/arka plan ayrıştırması, gürültüsüzleştirme ve gölgesizleştirme işlemleri sonrası belirlenmesi ve trafik ışıklarının akıllı kontrolu için kavşağın değişik bacaklarındaki doluluk analizinin nasıl yapılabileceği konuları işlenmiştir. Seçilmiş bazı video dizinlerinye yapılan benzetimler sonrası yöntemlerin başarıyla uygulandığını göstermek amacıyla makalede örnekler sunulmuştur. Gölgesizleştirme işlemi hiçbir zaman 100 % doğru olmadığı için ön plan görüntüsünde hareketli nesne diye belirlenen bazı bölgeler yanlış kabul çıkmaktadır. Bu yüzden şerit doluluk oranlarında 5% lik bir hata payı olabileceğini kabul etmek gerekir. Projenin devamında ayrıştırılan ön plandaki nesnelerin özellik çıkarımı, nesne sınıflandırma ve kırmızı ışık ihlalleri üzerinde araştırmalar yürütülecektir.

7. Teşekkürler

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